HYPERLOOP
GLOBAL

POLYLOOP
MONTREAL

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1 General

1.1 About the team

PolyLoop is a Hyperloop design team from Polytechnique Montréal founded in 2020 and currently composed of over 50 members. The team is divided in 11 sub-teams managed by multiple technical and non-technical leads as shown in the diagram below.

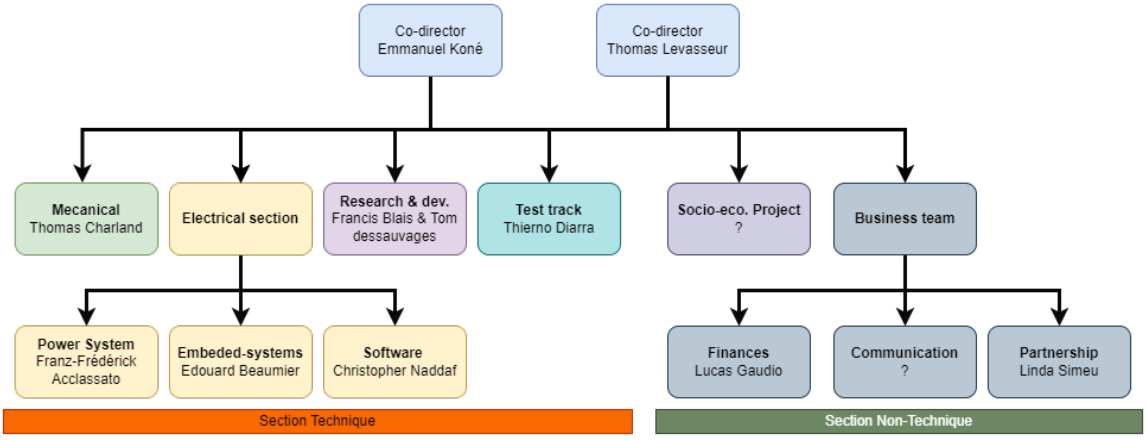

Fig. 1: PolyLoop organizational chart

The team went through a lot of ups and downs during its history but always managed to design and present impressive prototypes at the Hyperloop Global events. PolyLoop won the structure and aeroshell award 3 times, and we are ready to conquer many more in the following years.

1.2 Members List

Firstly, we would like to thank all of PolyLoop's members for their participation in the writing of this document.

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Jahed, Karim
Sam, Owen
Chaghouri, Chrystelle
Roy, Vincent

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Simeu, Linda Augustine
Sitbon, Célestin
Diallo, Boubacar Siddy
Abbas, Charbel
Sebatigita, Elsa

Administration

Thiemele, Gilles

Fig. 2: PolyLoop team

2 Subsystems description

2.1 Structures/Chassis

Overview of the design

This year again, our team will design the pod's structure based on structural design concepts. It will carry previous Hyperloop structure features such as a frame made out of 1-inch square aluminum tube.

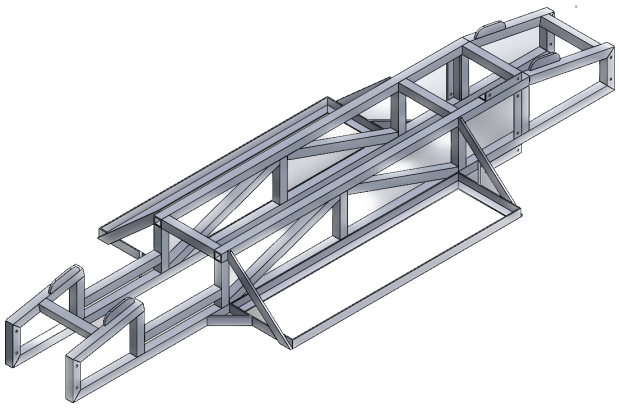


Fig. 3: Typical aluminum structure

Structural design is a methodical investigation to get the economical specification of a structure or a structural element to carry the predicted load safely. With the application of structural design, we can obtain the required size, grade, reinforcement, etc. of structural components to withstand the internal forces calculated from the structural analysis.

Modelling

That being said, we start by modeling the desired structure's skeleton, such as its foundations, frames, and other elements. It is essential for it to anticipate loads since it will be the basis for the whole process. We need to consider the maximum mass of 236 kg and the maximum length of 3 meters for the main pod. It will be designed symmetrically over the rail to have a center of mass that lies on the track.

Load Analysis

Next, one must identify all of the possible loads that the structure may experience during its lifetime. Moreover, these loads pose threats to the structure's purpose. Usually, this refers

to forces, torques, pressure, etc. In real life, different loads act simultaneously with the structure; hence, to accommodate this, load analysis also involves creating load combinations depending on the standards used. For instance, a simultaneous action between dead load and live load can be a load combination.

Our structure is first loaded under its own weight. Then, the acceleration applies another set of forces. There are also friction and drag forces that oppose the pod's movement. Finally, vibrations along the ride could bring the structure to resonance. The forces acting on the pod are described in table 1. Also, we need to consider critical situations such as sudden stops from top speed.

Tab. 1: Forces acting on the main pod

Force	Details	Value
Pod Weight	$100\,kg$ (All component included)	$981\,N$
Pod Acceleration	$0.93\,m\,s^{-1}$ (Assuming constant acceleration)	$93\,N$
Gearbox Friction	Force generated	$39.763\,N$
Pod Drag Force	$10\,m/s$ (Compute from ANSYS fluent)	$13.14\,N$
Sudden Stop	Conservation of Momentum	$9 \times 10^5\,N$

Structural Analysis

After identifying such loads comes structural analysis. The goal here is to investigate how our model behaves with different load combinations. There are different methods to analyze a structure and would depend on a lot of factors. A full analysis of a structure usually entails identifying the internal effects of different load combinations (normal, shear, moment, and torsion stresses), and the deformation/deflection of the structure. As a result, the observed effects would be the basis for designing the individual elements of the structure. We then use Ansys to calculate the various load effects to assure a minimal safety factor of 2.

Design

Structural design is perhaps the most important step in the process. It involves dimensioning the different parts of the structure based on the results of the analysis. Also, it is when choosing the right orientation, section, and other details that are relevant to the structure comes into play. For proper design, this step depends on whether strength-based or performance-based philosophy is applied.

We will use 1-inch square aluminum tube 6061 T6 in this frame design because it is much lighter than other metals. It has very high mechanical strength levels and that is why it is so broadly used in aeronautics, a very demanding field. Also, it is very resistant to many forms of corrosion and has almost zero maintenance cost. It also has the property of not being ferromagnetic, an important feature considering the use of magnetic fields in our overall design. The whole is also welded with 4043 aluminum wire .035" in diameter to solidify the assembly. Welds are stronger than nuts and bolts or rivets and does not loosen with vibration. The tube's thickness is set at 1/8 inch considering our welding equipment and our welding experience. Welding thinner than 1/8 inch becomes harder with our equipment. However, other components added to the chassis will be firmly fixed to the aluminum chassis.

An initial modal analysis revealed flaws in our design. Both ends of the chassis were loose and fully distorted at different vibrating frequencies. The eighth mode is shown in Figure 4 at 48.209 Hz. This design is thus non-viable.

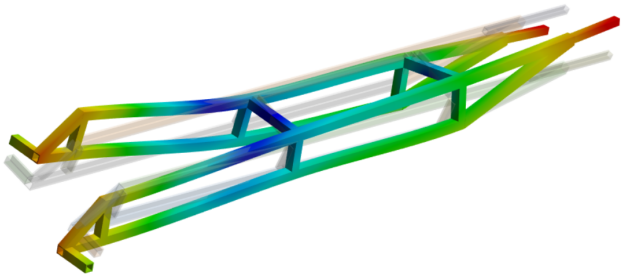


Fig. 4: Not to scale chassis deformation representation at 48.209 Hz

We have modified this design several times to achieve a frame that will not distort as much as an equivalent frequency. We added a rod on both ends to counteract the displacements observed in the figure 4. At a frequency of 43.835 Hz we observe a max displacement of 16 mm at the red spot in figure 5 compared to 837.45 mm in figure 4. Thereby, the structure will resist the vibration and the different modes encountered. The main results of the modal analysis are presented in appendix A.

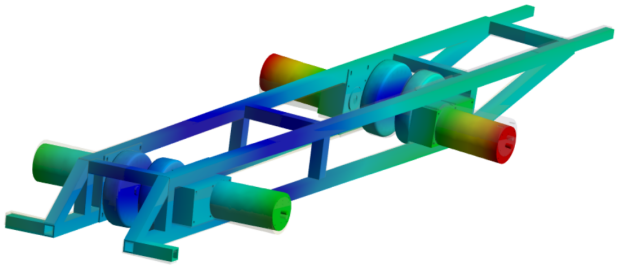


Fig. 5: Not to scale chassis deformation representation at $43.835\,Hz$

Following the dynamic analysis, we apply static load to conclude on the rigidity. We first test the situation where two people lift the pod at both ends. The gravity exerts a certain force. We also add another $30\,kg$ of material on top (general and over estimated). The maximal deformation is about $0.38137\,mm$ and shown in figure 6 using a 27 scale factor. The equivalent stress at this spot is $9.3981\,MPa$. The safety factor is well over two in that case.

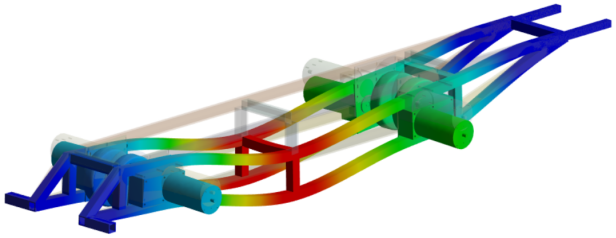


Fig. 6: Deformation under gravity (27 scale factor)

The last case scenario refers to a sudden stop from maximum speed. Imagine the pod hitting an obstacle sticking out of the track, causing a sudden stop. The front of the pod will feel a force equivalent to $914\,000\,N$. The max strain felt by the pod will be around $7\,742.9\,MPa$ with a maximum deformation of $100.35\,mm$ (Fig 7). We conclude that the structure will not survive such an impact. There would be permanent damage at least.

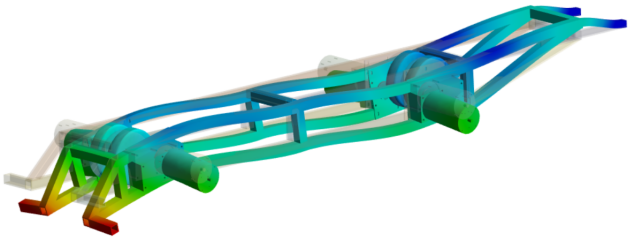


Fig. 7: Deformation under sudden stop ($914\,000\,N$ of force - true scale)

To conclude, the structure easily resists its own weight and over. It will certainly resist the pod's acceleration ($0.93\,m/s^2$). Moreover, the added rod helps counteract vibrations. We are not designing a tank, so we do not want to resist exponential forces. Therefore,it will most certainly not resist a full-speed impact.

Timeline of the design process

We are currently finalizing the design phase. We have spent the last months working mainly on other subsystems. In the past few weeks, we did the modelling and the load analysis. We did a first SolidWorks model and performed a structural analysis using Ansys identifying the internal effects of different load combinations. We are currently working on various configurations to explore the best possible outcome. From there, we will do welding tests and then build the final product.

Faced challenges

The first challenge we faced was a timeline issue. We did not have time to carry out a complete analysis of other frame material possibilities. We used the aluminum tubing concept developed for the last two pod iterations.

To go lighter using the same material, we would need to change the fixing method. Our welding equipment does not allow us to weld thinner than 1/8". We should also limit the free space in the pod. We designed it to be able to gradually include all subsystems separately. Indeed, each subsystem has more space than required to ensure it fits.

For the next iteration, we should entirely redesign the structure, perhaps choose another material that fits all our requirements, and plan where all the components will be more precisely.

2.2 Braking

Our team want to improve its braking system. Having the frictionless Hyperloop concept in mind, we are leaning towards a magnetic brake system. This option ensures the safety of the demonstration without damaging the track. Those brakes are not in contact with the rails, thus no scratch or damage can be induced as well as no instantaneous pressure of more than 34 MPa at any point. Moreover, because it is distant to the beam web, it will be tolerant of all potential variations and defects in the track.

Many possibilities are available. For starters, we need to choose between permanent magnets and electromagnets. Then we need to analyze different acceptable braking configurations and finally demonstrate that they will not overheat the track past 70 °C.

Permanent magnets or electromagnets

One of the many challenges of designing a magnetic braking system is to make sure the magnetic field of the magnet is close enough to the conducting material. This ensures the induced current is properly generated and the braking force generated is enough to slow down the pod. Each proposition comes with their advantages and disadvantages.

Electromagnets First and foremost, an electromagnet has an electromagnetic field that depends on the current flowing inside it, which means it can be adjusted to our needs. It is also less strict with the distance required between it and the rail due to the variable magnetic field. However, it also comes with potentially grand needs for power. The power could either be shared with the batteries of other subsystems or it could have its own set of batteries. Sharing power with another subsystem could inhibit that subsystem, making it inefficient. Having its own battery set will add some mass to the pod and could make it harder to move.

Permanent magnets Furthermore, permanent magnets have, unlike an electromagnet, a fixed magnetic field, meaning the distance between it and the rail could be more restrictive and it could still affect the trajectory of the pod even if it's some distance from it. The challenge here is to move the magnet close enough to the rail to induce enough current and keep it from affecting the pod by moving it further from the rail or completely block the magnetic field from reaching the rail. There is a way to increase the intensity and the reaching distance of the magnetic field of permanent magnets using a Halbach array. A Halbach array is composed of five magnets in a specific configuration as shown in Fig. 8. Their magnetic field is all aligned in a specific way. This configuration makes the magnetic

field more intense on one side and almost nonexistent on the other. It allows a greater gap between the array of magnets and the rail without losing any braking power. Still, this solution doesn't solve everything because we still have to move the magnets even further than before or block the magnetic field.

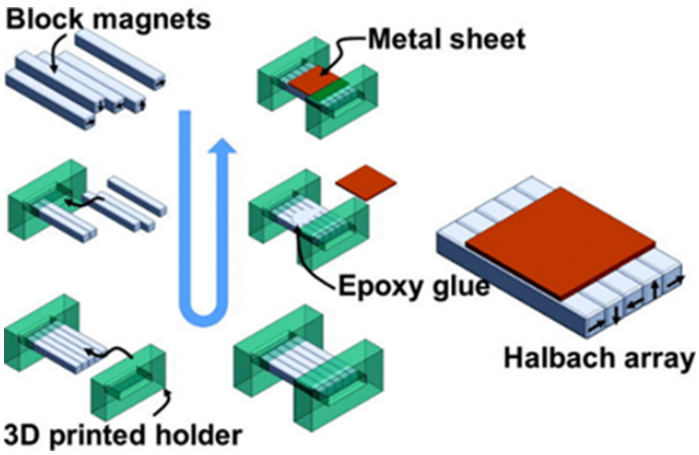


Fig. 8: Halbach array permanent magnet configuration

Analysis of Halbach array

To reinforce our understanding of magnetic brakes, we ran simulations using COMSOL. Our goal was to get a better comprehension of a Halbach array and have some data to support our final decision. We started by trying to mimic the exact set up we would have when it will be build, shown in appendix A in Fig. 55. The configuration shown is optimal, it is the one that puts the brakes to what we can theoretically do with our current means, about $5\,mm$ from the rail. There is only one part of the brake because we wanted to get a better understanding first, rather than have even more components and no understanding at all. A more realistic approach is scheduled. The simulation results are dependent on the velocity which has a range of $0\,m/s$ to $30\,m/s$. Fig. 56 in appendix A shows a peak around $6\,m/s$, and then the braking force suddenly lowers as expected. It is optimal for our pod as the highest braking forces are in our range of speed. One thing to note is the values of the forces in this graph are in N/m. We simply need to multiply each value by the width of the Halbach array to get the actual braking force. For instance, the highest braking value obtained is $26\,777\,N/m$. Multiplying by $0.03\,m$, the width of the Halbach array, gives a braking force of $803\,N$.

Finally, our team decided to choose the Halbach array with permanent magnets. This solution requires much less power from batteries, which was the main issue of the electromagnet. We also have data from simulations to backup our decision.

Braking configuration

The Halbach array requires a total of five magnets as mentioned earlier. As shown in the Rulebook, the arrangement of the braking system must be symmetrical, meaning we must use a minimum of two Halbach arrays, so a total of 10 permanent magnets. The ones used in our system are neodymium N52 magnets. Those can be quite expensive so we can't use as many as we want. This limits us in the configuration but it isn't an issue.

The five magnets are touching each other, shown in Fig. 9, to maximize the efficiency of the Halbach array. Usually, a Halbach array is made of squared magnets. However, in our case, the ones that we have are rectangular. This changes the resulting braking force. It is an insignificant difference in our case. They are fixated on an aluminum plate with aluminum bolts which makes the handling easier as aluminum is non-magnetic. The complete assembly is shown in Fig. 10. The system has a lock to maintain it downward while we don't need to brake. We can see it in Fig. 11. It retracts to let the array go up to brake. The system is failsafe as the lock will retract automatically when there is no more current in it.

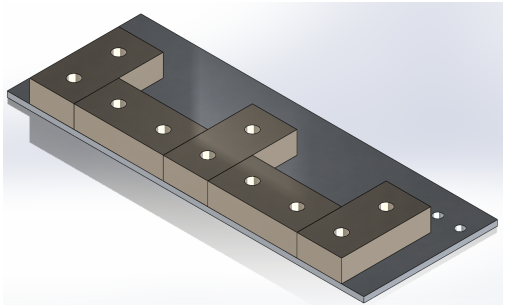


Fig. 9: Design for the Halbach array arrangement with completely visible magnets

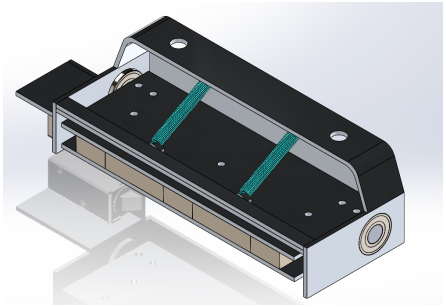


Fig. 10: Complete design for the braking system

We decided to put one Halbach array on each sides of the track shown. Fig. 11 shows one half of the placement.

Challenges met

During the conception of the magnetic brake system, the team dedicated to the magnetic brakes had very little knowledge on the matter. Also, it is very uncommon for a magnetic brake system to take the form shown in Fig. 11. Usually, there is a fixed rotating disc that is used as the conducting element in the magnetic brake system, usually in heavy vehicles. Having close to nothing to start with was proven quite challenging, but as seen earlier, we've found a functioning design. Also, levitating land vehicles aren't common, which makes their braking technology quite inaccessible, making it harder to design our brakes. Another

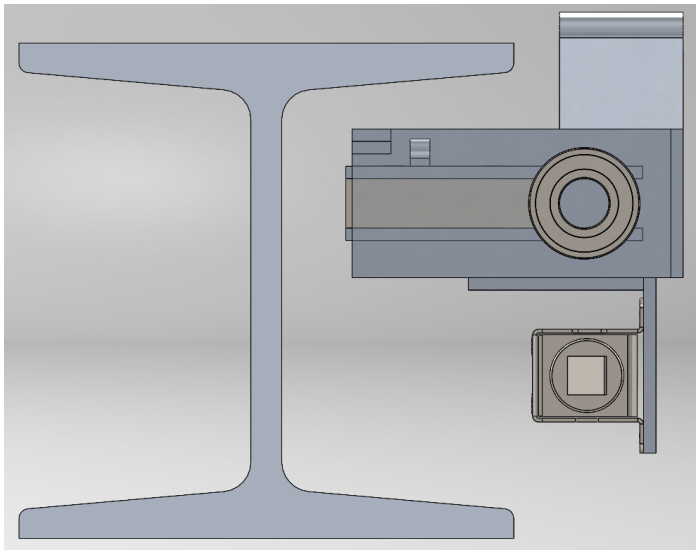


Fig. 11: Placement of one half of the braking system

challenge was to choose between the electromagnet and the permanent magnet, as they both have interesting characteristics. It took a couple of months of reading and debating to choose the permanent magnets.

Timeline of the design process

Back in September, the different mandates were assigned to the different members of the mechanic team and one of the was the conception of magnetic brakes. Although some ideas and prototypes were there before, making it a reality and implementing it in the pod was something never seen before in Polyloop. It was only in late December that we made the decision to stick with permanent magnets due to the results of the simulations and the knowledge we got from them. The design shown in Fig. 11 was produced some time afterwards. We still have to find a way to nullify the effect of the magnets. Then, there will be a testing phase that is to come after building the brake. Afterwards, they should be operational.

Emergency brake

Although the magnetic brake should act as a fail safe system, our speed reducer can also bring the pod to stop. Our motors are connected to Standard Speed Reducers shown in figure 12. A single stage of roller chain reduction is generally considered to be about 98% efficient. Add in some friction from the five bearings, and we are at about 90% efficiency.

Fig. 12: Opened Ampflow standard gear speed reducer

The friction in the speed reducer is about $40.99\,N$, thus, the total friction force is $F_f = 163.96\,N$. In case of total power failure, it will be capable of stopping the pod. Indeed, the pod wheels are always in contact with the rails and always apply the same friction force. It also brings the pod to a complete stop after the magnetic brake effect. Those magnetic brake has a braking force proportional to the relative speed, thus it can't theoretically stop the pod.

We obtained the friction force by testing the motor. We rolled a string around the wheel to which we hung a mass. Then, with an high-speed camera, we determined the time the mass traveled a certain distance. We then summed the force to obtain the friction force for a single-speed reducer. From the Appendix A, we calculate the deceleration and the time to immobilization. From there we can calculate the approximative distance traveled by the pod in case of total failure. Indeed, by using simple kinematic equations and hypothesis, we can get the distance traveled with quantities associated with the motion of objects featured in Tab 2. With the calculation in appendix A, the pod will travel $26.27\,m$ before complete immobilization in total power failure.

Tab. 2: Quantities associated with the motion of the pod

Variable	Value
Deceleration Time	$\Delta t = 5.7464\,s$
Nominal Speed	$v_0 = 9.14\,m/s$
Final Speed	$v_1 = 0\,m/s$
Deceleration	$a = -0.409\,m/s^2$

2.3 Aeroshell

Overview of the design

After winning the Aeroshells Award three years in a row, we we'll maintain the same aeroshell as the last competition and focus on other main parts. We decided to lean toward the original sketch (Fig 13) designed by Polyloop at their founding in 2020. At that time, manufacturing those ideas was too complex, it required composite knowledge.

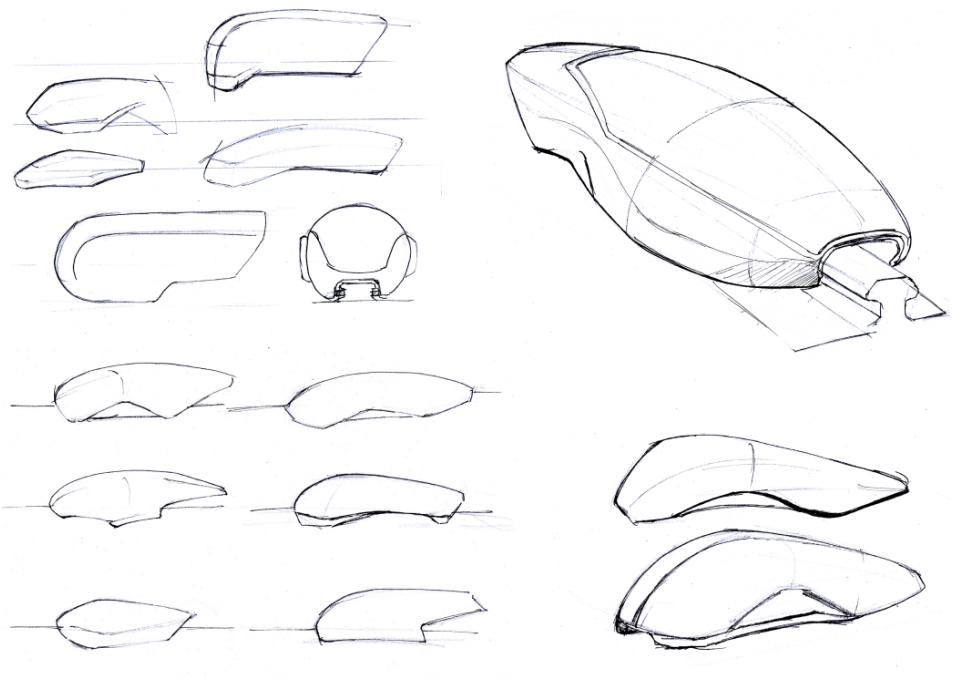


Fig. 13: Exploration drawings

In 2023 we launched our first carbon fiber aeroshell. The final product had some issues that will be resolved on this year's pod. First, the last iteration was made out of three parts; two bottom parts and an upper one (Fig 14 and 15). We had problems while trying to fit all those parts together. During the manufacturing process, we did not ensure the fitting between those parts. Both the bottom and the top parts did not have a good finish which made the alignment impossible.

Moreover, it was not practical. Once the upper part is fixed, we could not access the interior of the pod to make some tweaks. Lastly, the final pod rendering was poor quality. The surface epoxy was rough and unacceptable.

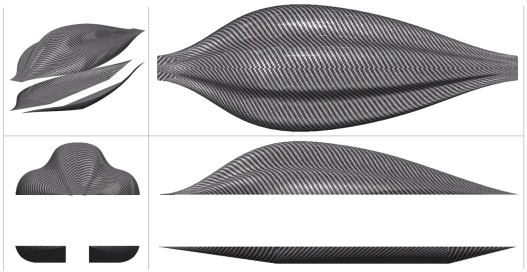


Fig. 14: Last year aeroshell CAD

Fig. 15: Last year aeroshell rendering

Solution

To overcome these problems we decided to design a monocoque aeroshell using the exploration drawings of Fig 13. We will also cut out a hood into this monocoque to ensure the assembly of these two pieces. They will be joined by hood pins to facilitate the manipulation.

Fig. 16: CAD of this year iteration showing
the main body and the hood

It will ensure the pod's protection against wind, dust, and particles. We have also carried out preliminary finite element analysis to conclude that this is not only appealing to the eye but will ensure excellent aerodynamic properties at the maximum operational speed of the pod.

Computational Fluid Dynamics (CFD)

First, we started with modeling the aeroshell in SolidWorks using 3D splines to design the aerodynamic shape (Fig 17). We added other features such as cuts, shell, loft, etc (Fig 16).

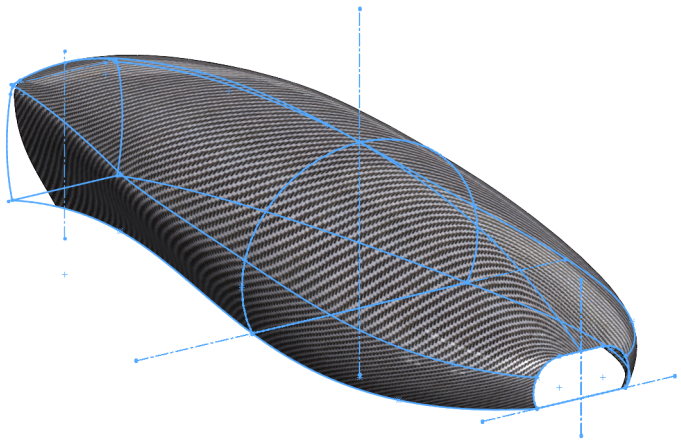


Fig. 17: SolidWorks model resolving the 3D sketch

Then, we used ANSYS Fluent to compute finite element analysis on the aeroshell in order to determine the drag coefficient, the drag force, and the lift and to see the overall behavior of the shell in certain conditions. The air becomes quickly turbulent. Because we have a Reynolds number superior to 10^6 we are in the presence of a turbulent flow. To this conclusion, we use a $k - \epsilon$ turbulence model because it converges more easily and is widely used in industries. It is one of the Reynolds-averaged-Navier–Stokes (RANS) models that is mostly used to simulate mean flow characteristics for turbulent flow conditions. It is a two equations model that gives a general description of turbulence utilizing two transport equations.

Without all the elements inside the pod, we still wanted to represent a realistic situation. We added two rail elements to our aeroshell CFD simulation to see their interaction with the flow. In order to get the desired mesh quality, we had to look at a bunch of parameters. We started by applying a tetrahedrons method to our mesh. Then we tweaked faces and body sizing elements to obtain the right Element quality, Skewness, and Orthogonal Quality (Appendix A). The mesh with the worst quality is located at each vertical side of the rail. Fortunately, these sections are not studied and their mesh quality doesn't bother the rest. Moreover, the general meshing has a skewness inferior to 0.7.

We also added an Inflation of 16 layers with a growth rate of 20% to well capture the pod boundary layer (Fig 18 and 19).

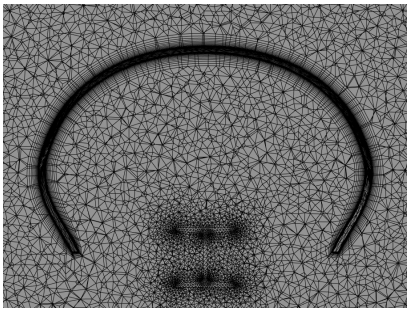


Fig. 18: Cross section of the meshing - distant view

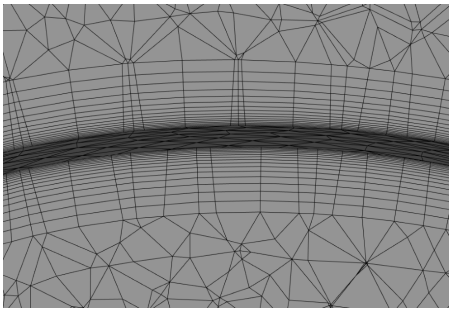


Fig. 19: Cross section of the meshing - close view

After the mesh quality, we needed to analyze the $y+$ dimensionless parameter. On the pod's boundary layer, we obtain $y+$ value between 0.21 and 7.83 (Appendix A), almost inside the range recommended in the literature ($1 < y+ < 10$). On the other hand, the $y+$ value on the wall with the no-slip condition should be between 30 and 300, and our analysis gives us values $77.54 < y+ < 203.27$ (Appendix A). This means that the $k - \epsilon$ turbulent model will compute better in the mean flow than in the boundary layer as intended by choosing this model. Although, considering the precision needed, it is enough accurate to consider the results of this analysis. The scaled residuals converged under 0.001 with only 263 iterations using second-order solution methods (Appendix A). Moreover, the Mach number is relatively low, $M \leq 0.3$. At that number, the Mach number is unimportant compared to the Reynolds number.

For a velocity of $10\,m/s$ we get a drag coefficient $C_d = 0.12904425$(A), and drag force $F_D = 7.9039601\,N$(A). Also, the lift force is $F_L = 1.4687748\,N$(A) given by the lift coefficient of $C_l = 0.023979997$(A). The gauge pressure is drawn in Fig 20. It reveals that the pressure is pretty constant around the pod. The capsule will ensure good aerodynamic properties during the run while protecting the pod from the environment.

Timeline of the design process

From September to November, we designed and analyzed the aeroshell shown in figure 16. We started machining small prototypes in foam to eventually machine a mold to apply carbon fiber in 2024. Meanwhile, in January 2025 we plan to test vacuum bagging carbon fiber on smaller prototypes. Moreover, we are in discussion with technicians to test one of

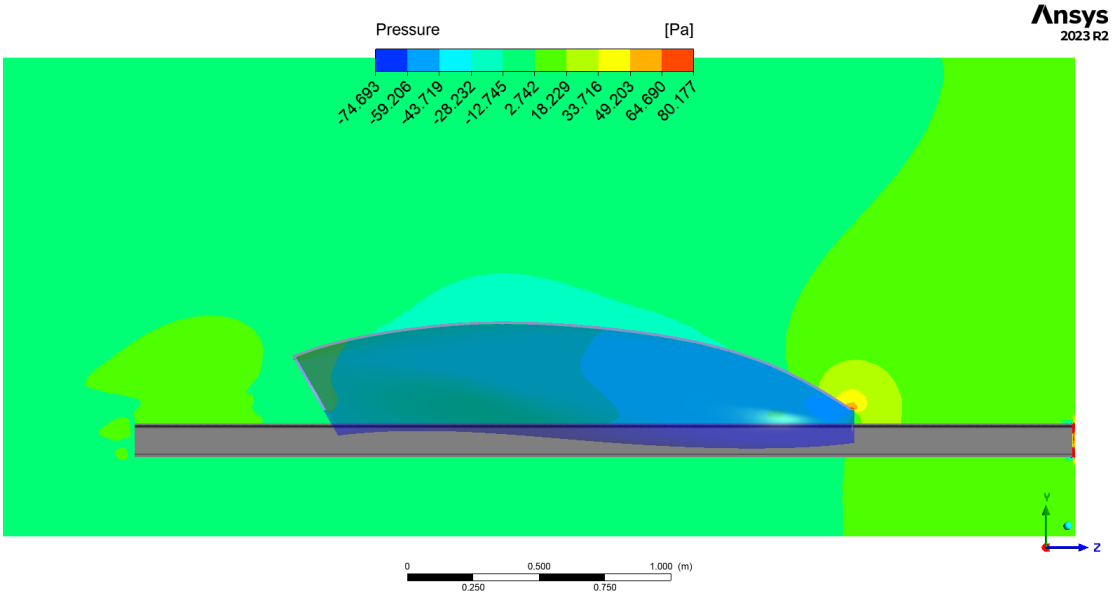


Fig. 20: Gauge pressure around the pod

those prototypes in our school wind tunnel. Then we will build the final full-scale prototype.

Challenges met

We faced a wall when we had to find a way to fix the aeroshell to the frame. We successfully found a clever way to use a swivel ball joint with a flat base mounted inside the pod. Another challenge concerns the calculation time of CFD analysis. To have precise results, you need an excellent mesh, but in exchange, it requires more calculations. After a simulation, we changed our design and had to recalculate the analysis again and again. This could take up to 24 hours.

2.4 Cooling Systems

Context Setting

The recent competitions revealed an issue with the circulation of heat flow of the pod. Indeed, exposure to direct sunlight at its zenith in temperatures exceeding 30 degrees Celsius weakened the external structure. Our structure is made of carbon fiber composite solidified by epoxy. Some areas on the structure had softened due to the degradation of the optimal properties of the carbon fiber and epoxy mixture. Additionally, the heat absorbed by the shell was such that handling it could cause mild burns to the fingers. Also, since our technical team does not yet possess mastery of a Linear Induction Motor (LIM) requiring 25,000 W of electrical power, the team will introduce four direct current (DC) rotary motors on wheels to substitute for the main propulsion system until the LIM technology is mastered.

These DC motors are connected to speed reducers to increase torque. Recently, a short no-load test on this arrangement revealed an operating temperature of over 50 Celsius degrees for the reducers. This thermal gain over a short period raises questions about the need of an internal cooling system. Therefore, a reassessment of the current material and/or coating of the external structure, as well as the implementation of a cooling system, is to be expected. Cooling systems of the pod will be separated into two big sections which are: Cooling of the speed reducers and Cooling of the aeroshell.

Cooling of speed reducers

Proposed Methodology :
1. The team will assess the current issue:
a) Evaluate the heat-related problems on various components of the capsule.
b)Examine the impact of heat on the aluminum of the speed reducer.
c)Determine material and environmental characteristics.

2. The team will identify the characteristics of current materials and the environment (conductivity, absorptivity, conduction coefficient, convection coefficient, etc.). The key will be to formulate appropriate assumptions regarding initial temperatures.

3. The team will then conduct Heat Transfer Analysis using EES (Engineering Equation Solver). The heat transfer will be evaluated on the speed reducers to determine heat transfer characteristics.

4. From the results of the previous step, the team will propose cooling solutions based on

preliminary analysis. An example of a solution could be as simple as adding a fin system that uses incoming airflow to cool down the pod.

5. When the solution is chosen, the team will validate numerically with EES that the chosen solution effectively regulates the system temperature.We will also extend the analysis to a 3D Model so that we can finally use Ansys Workbench and Steady-State Thermal Function.

Cooling of aeroshell

Proposed Methodology
1. The team will assess the current issue :
a)Evaluate heat-related issues on various components of the capsule.
b)Determine the glass transition temperature (Tg) of epoxy.
c)Determine material and environmental characteristics.

2. The team will identify the characteristics of current materials and the environment (conductivity, absorptivity, conduction coefficient, convection coefficient, etc.). Again, the key will be to formulate appropriate assumptions regarding initial temperatures.

3. Then, the team will conduct Heat Transfer Analysis using the EES (Engineering Equation Solver) of the shell considering radiation and convection. From the heat transfer results, the team will propose solutions based on preliminary analysis such as introducing a new material with lower absorption and higher refractoriness.

4. The team will reanalyze the chosen solution with EES to validate.

5. Team will validate numerically with EES that the chosen solution effectively regulates the system temperature. We will also extend the analysis to a 3D Model so that we can finally use Ansys Workbench and Steady-State Thermal Function.

Conclusion

It should be noted that the solutions chosen for these two cooling systems will be studied and presented during the current semester as part of the capstone project of two members of the mechanical team. The chosen solution could be integrated into the final module depending on the progress of these two synthesis projects.

2.5 Guidance

Lateral guidance

The lateral guidance and suspension system (Fig. 21) consists of four fixed wheels positioned on both ends of the frame and on either side of the track. The 50 mm wide tires are made of high-quality solid rubber (hardness 80 shore A) which will ensure both surface preservation of the track and smooth rolling. The assembly also includes Belleville washers and rubber washers to absorb vibrations during operation.

By trial and error, we found the ideal number of rubber washers to put on each bolt to obtain a good mechanical fit with the track.

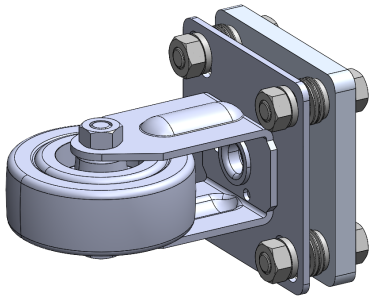


Fig. 21: Lateral guidance and suspension

The lateral suspension system uses a rubber washer to absorb the vibrations of the pod. They are only useful in compression; they don't have any influence on the pod when they are in traction. Thus, with CES Edupack, the compression properties of the material that constitutes the washer have been evaluated.

Stiffness of the spring

In the next few formulas, a unit factor will be used between the movement of the wheel and the spring. To respond to this uncertainty, a security factor of three will be used. To find the necessary spring stiffness, two factors are important: the energy needed for a smooth movement of the capsule and the force needed to realign the pod. For the first factor, we suppose a conservation of the angular moment:

$$I\omega_i + mv_i(r-h) = I\omega_f + mv_f(r) \tag{ii}$$

Where v_i and ω_i are the initial linear and angular speed of the wheel, v_f and ω_f are the final linear and angular speed of the wheel, m is the mass of the main pod, r is the radius of the wheel, h is the max misalignment possible ($10mm$) and I is the inertial

moment of the wheel ($I = \frac{1}{2}mr^2$ for a disk)

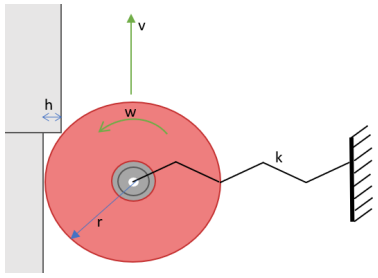


Fig. 22: schematic

With (ii) and assuming an initial speed of $30\,km/h$, a radius of $25\,mm$ and a mass of $100\,kg$, it is intuitive to find the final velocity of the wheel:

$$v_f = 25\,km/h$$

With this new value, with the energy conservation law, it is possible to find the minimum stiffness k needed to move up the wheel without the wheel jumping out of the track with a security factor of 3, the stiffness needed for such a speed differential is $k = 127\,000\,N/m$.

$$\frac{1}{2}mv_i^2 + \frac{1}{2}k\Delta x^2 = \frac{1}{2}mv_f^2$$

For the second factor, Since the spring need to be at half its maximal deformation when it is stable and we don't want it to be completely compressed the force of the spring at $\frac{3}{4}$ of its travel should be stronger than half the friction force plus the force of the spring on the other side of the track which should be at 1/4 of its travel.

$$\begin{aligned}\frac{3}{4}k\Delta x &= \frac{1}{4}k\Delta x + \frac{1}{2}(mg\mu)\\ \frac{1}{2}k\Delta x &= \frac{1}{2}(mg\mu)\\ k\Delta x &= mg\mu\end{aligned}$$

Since we need to know the maximal travel of the spring, we can't have the value of the constant needed to realign the pod.

The washers are made of polyester fiber, most specifically of SBR unreinforced. This material as the property of a compression set of 5% to 10% at $23°C$ and 10% to 20% at $70°C$. This means that for lateral suspension, at $23°C$ the maximum compression would

be 1.27 mm and 2.54 mm at $70°C$. Knowing that the temperature in the pod would be between $23°C$ and $70°C$, the rubber will be perfectly able to resist the vibrations. We then stack washer to obtain our desired stiffness.

Vertical guidance

The set of 4 propulsion wheels discussed in section 2.7 is used as the vertical guidance. Indeed, the pod will not levitate, thus there is only a down force to counter due to gravity. The set of wheelmotor are equally spaced in pairs (Fig 24), thus sharing the load. This configuration assures stability.

2.6 Electronics, Controls, and Communication

State Machine

Our pod's behaviour is controlled by a state machine in order to guide it in the order of events that must be followed for a successful operation. As shown in the figure below, the pod starts in an Idle state where it awaits human confirmation to undergo a series of safety checks before giving its approval for launch. If one of these checks fail, the pod immediately enters an emergency state corresponding to the specific issue. Otherwise, an operator can launch the pod which will start its acceleration phase until it reaches the appropriate distance to coast. Once the pod reaches the maximum coasting distance, it will start to brake and come to a halt, followed by a transition to its initial state. During the entirety of the pod's movement, safety checks (composed of the same tests as the pre-launch checks) will be running in the background, and if at any moment, a test fails, the pod immediately enters the appropriate emergency state and activates its brakes. It is also not shown on the figure, but if an emergency stop button is activated, the pod will immediately enter a general emergency state in which it will activate its brakes, shut its main power down, and switch its LED interface to a solid red colour.

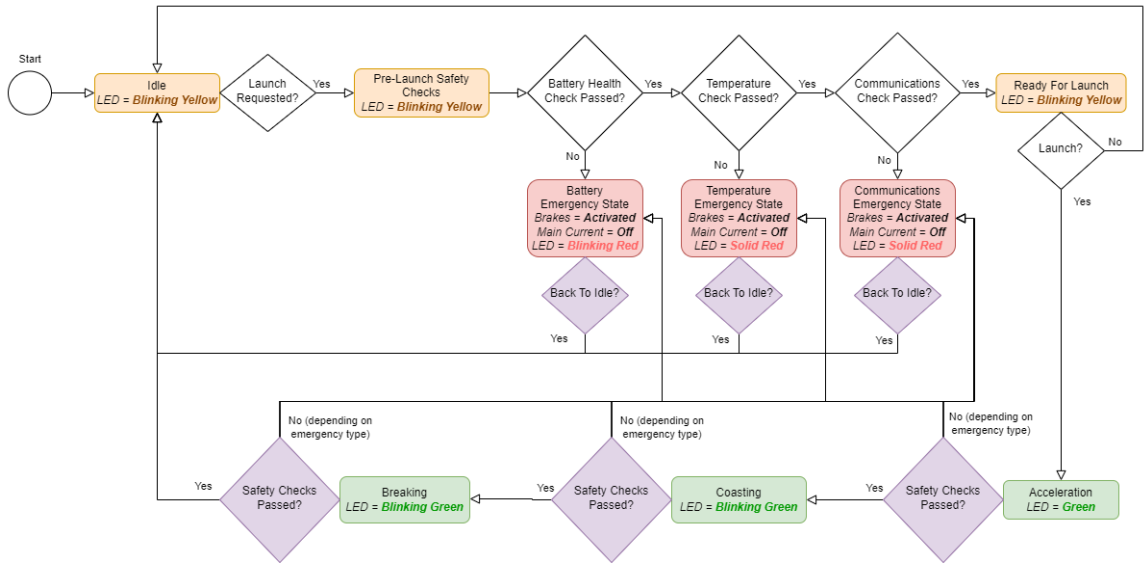


Fig. 23: Pod State Diagram

Last year's state machine was programmed in Python since we were using a Raspberry PI 4 as the main computer of the pod. However, this year we are implementing the design differently by programming it in C++ to fit a new architecture that implements multiple ESP32 microcontrollers (see the Embedded Systems Overview section below). The pod will

also be controlled remotely via a Web App running on an external computer that will allow operators to see key information such as the pod's current state and telemetry in real time.

Embedded Systems Overview

The electronics on the pod are designed with custom printed circuit boards (PCBs). By creating custom PCBs, we can adapt the circuitry to our specific needs, ensuring optimal performance and seamless integration of all components. The team is relatively new to PCB design, so we have focused on making the designs simple yet effective to ensure reliability while we build our expertise.

The system controls are divided into three different types of circuits:

- **Motor Control Circuit**: This circuit controls the motor speed and brakes. It includes onboard sensors for speed and acceleration of the pod, allowing control over the pod's movement.
- **Battery Monitoring Circuit**: Connected in series with the batteries, this circuit monitors the system for any anomalies using current, voltage, and temperature sensors. It also includes fuses and relays to enable easy shutdown of the system in case of issues, enhancing safety.
- **Communication Circuit**: This circuit enables communication between the pod and a base station, which will be an external computer. The two-way communication allows us to transfer data in real time and to start up or shut down the system at any given moment, providing flexibility and control.

All boards use an ESP32 microcontroller (MCU) due to its versatility and built-in wireless communication capabilities. The subsystems communicate internally via the Controller Area Network (CAN) bus, and the pod communicates with the base station using the ESP-NOW protocol.

Systems Architecture

The system is divided into four subsystems, each serving a different purpose in the pod:

Lys (x2)

Functions:

- Processing key information such as status and speed.
- Controlling motors and brakes.

The Lys subsystems are critical for propulsion and deceleration. They process data from sensors such as accelerometers and optical sensors to determine the pod's current speed and acceleration. Using this data, the ESP32 MCU in each Lys board controls the MD25HV motor driver to adjust motor speed and activates relays for the braking system when necessary.

Oranger (x2)

Functions:

- Monitoring the state of high-voltage batteries (current, voltage, temperature).
- Shutting down the system using relays in case of problems.

The Oranger subsystems are responsible for the safety and management of the high-voltage batteries. They continuously monitor battery parameters using voltage sensors, current sensors, and temperature sensors. If any anomalies are detected, such as overcurrent, undervoltage, or overheating, the system can initiate a shutdown using relays and fuses to prevent damage or hazards.

Orchidée (x1)

Functions:

- Monitoring the state of low-voltage batteries (current, voltage, temperature).
- Acting as a CAN hub for information transmission.

Orchidée monitors the low-voltage batteries that power the electronics and control systems. It ensures these batteries are functioning within safe parameters. Additionally, Orchidée acts as the central hub for the CAN bus, facilitating communication between all subsystems. The ESP32 MCU in Orchidée manages data traffic to ensure timely and efficient information exchange.

Pivot (x1)

Functions:

- Providing a user interface (screen, buttons).
- Local data storage using an SD card.
- Controlling status LEDs.
- Communicating with the external computer.

Pivot provides the human-machine interface for the pod. It features a screen and buttons for user interaction, allowing operators to view the system's status and to issue commands. Pivot also handles local data storage using an SD card, recording operational data for analysis. Status LEDs controlled by Pivot offer immediate visual feedback on the system's state. Communication with the external computer is handled via ESP-NOW, enabling remote monitoring and control.

Communication

The pod's communication architecture comprises two primary components: the Controller Area Network (CAN) bus for internal communication and ESP-NOW for wireless communication with the base station.

The CAN (Controller Area Network) communication protocol is robust and a standard in embedded systems and industrial applications. Designed for high-reliability data transmission, it is particularly well-suited for scenarios where multiple devices need to communicate efficiently and reliably on a shared bus. In our pod, the CAN bus ensures that all subsystems can exchange information effectively, maintaining synchronized operation across the entire system.

For our implementation, we are using the TCAN332DR as the CAN transceiver. The TCAN332DR is a reliable and reputable CAN transceiver that interfaces directly with the CAN controller onboard the ESP32 MCUs. With a required supply voltage of 3.3V, it integrates seamlessly into our design, matching the voltage levels provided by our buck converter. The TCAN332DR's compatibility with the ESP32 simplifies integration and ensures reliable communication between subsystems.

ESP-NOW is utilized to enable wireless interaction between the pod and the base station. Leveraging the ESP32's built-in ESP-NOW protocol, the pod can communicate with the external computer, facilitating real-time data transfer and remote system control. Our design includes a Web App that displays all key characteristics of the pod on the base station, providing operators with vital information at a glance.

ESP-NOW provides quick and efficient communication with minimal setup requirements and independence from external network infrastructure. It allows for low-latency, peer-to-peer communication without the need for a traditional Wi-Fi network, which is ideal in our case where rapid and reliable communication is essential. The protocol's simplicity and effectiveness align with our design philosophy of keeping systems straightforward and functional.

Interface

The interface components provide essential interactive functionality for operators inside the pod.

The screen connected to the subsystem's ESP32 microcontroller will display real-time data such as the pod's internal temperature and the batteries's voltage. It will also allow us to display any warning messages, enabling us to promptly respond to any issue. With the help of buttons, additional control features can be added within the interface to adapt it to our needs.

We have selected a 0.96-inch OLED display module connected via an I2C protocol as our screen. The display is small but sufficient to present all the necessary information. The use of the I2C protocol and its compact size simplifies the implementation to our system. The status LEDs are also controlled by the interface system and follow the specifications mentioned in the competition rulebook. These LEDs will provide immediate visual feedback on the pod's operational status.

Voltage Regulation

Proper voltage regulation is essential to ensure that all electronic components receive stable and appropriate power. The components of our circuit require a 3.3V power supply, but our battery provides 24V. To address this discrepancy, we utilize a step-down voltage converter (LM2596) to reduce the voltage from 24V to a stable 3.3V suitable for our components. The input's XT60 connector supplies the 24V from the battery, which is where the LM2596 is

implemented following the manufacturer's recommendations. The efficiency and abundance of resources led us to believe that this buck converter is the best for our case.

Battery Management

Robust battery management is essential for the safety, performance, and longevity of the pod's operation. The battery management system includes components such as fuses, relays, voltage sensors, current sensors, and temperature sensors, all working together to monitor and protect the battery systems. It is important to note that this system does not replace the existing Battery Management System (BMS) used during charging; rather, it operates in addition to the circuitry in place for charging, providing enhanced monitoring and protection during operation.

Fuses are installed in series with the batteries, providing protection in case of an overcurrent situation. A socket will be installed so we can select the appropriate current rating of the fuse based on the system characteristics. This flexibility allows us to tailor the protection to the specific requirements of the pod, ensuring safety without compromising performance.

Relays allow for controlled disconnection of power to certain parts, such as the motors. They are controlled by ESP32 outputs, utilizing the 24V supply for proper control of a relay rated for 48V DC. This setup enables us to isolate high-power components safely and efficiently.

Voltage sensors monitor battery levels to assess the state of charge and detect anomalies in undervoltage or overvoltage conditions. A simple voltage divider circuit will suffice to monitor the approximate voltage value when connected to an analog read input on the ESP32. This method provides a cost-effective and straightforward way to keep track of battery voltages in real time.

Current Sensors measure the flow of current and allow us to monitor power consumption. Using the ACS781LLRTR-100U-T, we can safely measure the current of each battery system, ensuring the adequacy of operation. This sensor can handle high currents and provides accurate measurements, which are critical for detecting overcurrent conditions and optimizing energy usage.

Temperature sensors monitor the temperature of the batteries, the motors, and the ambient environment. By using strategically placed thermistors on the batteries and motors, we

can ensure that our system does not overheat. In addition to that, we will use an integrated circuit such as the AHT20 with an I2C protocol to measure the ambient temperature of the pod. Monitoring temperature is crucial for preventing thermal runaway and ensuring that all components operate within their specified temperature ranges.

Pod Controls

The pod controls include components that manage the propulsion and braking of the pod. The key components include accelerometers, optical sensors, the MD25HV motor driver, and relays for the braking system.

Accelerometers measure acceleration forces acting on the pod, allowing us to monitor the pod's movements closely. We are using the MPU6050 due to its high accessibility, easy implementation, and high refresh rate. The MPU6050 integrates both an accelerometer and a gyroscope, providing comprehensive motion tracking capabilities. By interfacing it with ESP32 microcontrollers, we can obtain real-time acceleration and orientation data.

Optical Sensors measure the speed of the pod by detecting the rotation of the wheels. By using an infrared LED module, we can accurately measure the number of times the wheels rotate at any given time.

MD25HV Motor Driver controls the motors that propel the pod. Instructions are sent via Pulse Width Modulation (PWM) from ESP32 microcontrollers. The motor driver receives control signals to adjust motor speed and direction based on input from sensors and control algorithms.

Relays are used for the braking system to engage the brakes and decelerate or stop the pod safely. The same system of relays used to disconnect the batteries from the system will be employed for the braking mechanism. The use of sockets for relays allows us to easily adapt the relay depending on the braking system requirements. Manual override capabilities will be implemented for this relay as well, providing operators with direct control if necessary.

2.7 Propulsion

The pod will be powered by DC electric motors mounted on 6 inches wheels proportioned by a speed reducer to improve the torque, thus the acceleration (Fig 25). We thought about doing an hybrid propulsion using both friction wheel and contactless propulsion given by a Linear Induction Motor LIM. But the LIM ratio weight-thrust is too high to be added onto the main capsule. It won't help the propulsion, in fact, it might be a liability. That is why we are building a pod with DC motor and also working in parallel on a custom Linear Induction Motor.

Propulsion of the pod

The electric propulsion system works by friction. In fact, we opted for the use of wheels in contact with the rail propelled by DC brushed motors. Compared to a previous iteration, which had only one wheel powered by two DC motors, this propulsion is achieved by the arrangement of two pairs assembled in two overall. The choice of this configuration is based on the principle of stability, which from a practical standpoint refers to pairs. The mass of the capsule is thus distributed across the four motors, reducing the effort provided per motor.

Secondly, stability is achieved through the maximum spacing between two aligned wheels. By increasing the distance between two support points, stability is increased. The motors of the same pair are not connected axially, allowing each to react to track imperfections and vibrations independently to the other.

The choice of motors for this iteration is based on the energy calculations. The required power and torque were calculated for a two, four, six, and even eight motors configuration. According to the specifications of the different engines, our choice stops on the E30-400-48-G4. According to the manufacturer's performance tables (appendix A), the motor chosen has a torque of $4.77\,N \cdot m$ for $16\,amps$, which allows us to calculate the acceleration of the capsule by the sum of the moments at the point C (appendix A).

In an effort to minimize mass and go as fast as possible within the confines of the track, four DC brushed motors with a ¼ speed gearbox were chosen (Fig 24). The choice of this gearbox was based on an analysis of the distance traveled before reaching the motors' maximum speed. Increasing the reduction ratio certainly decreases the outgoing speed, but it increases the torque and therefore the acceleration. The current configuration of four DC brushed motors with ¼ gearbox as well as a six inch wheels, would allow one to go at a theoretical speed of $9.14\,m/s$ within the limits of the track.

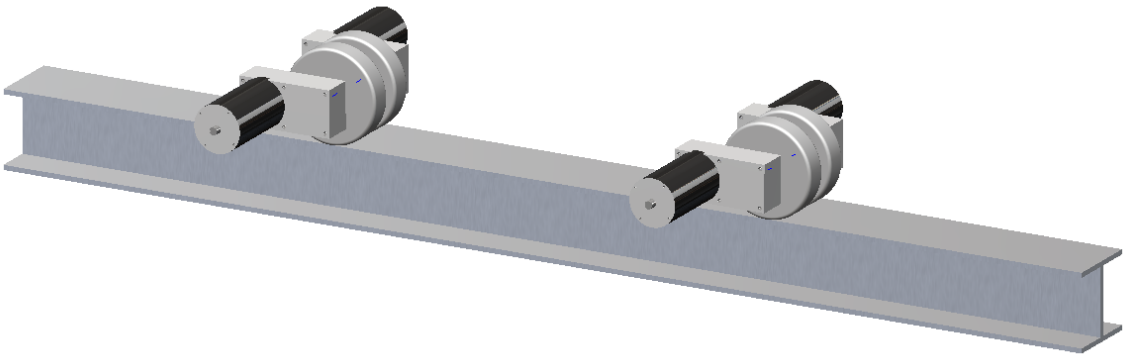


Fig. 24: Assembly of four gear-motor-wheel set in two pairs symmetrical onto the rail

Following the advice of different teachers, we bought an already existing set of motors, gearbox, and wheel. The reasons given by these professors included the complexity of achieving precision in constructing such a system.. We, therefore, purchased this assembly from *AmpFlow* (Fig 25), a manufacturer that offers quality machined products. An aluminum, machined gear box enclosure will certainly prevent part from dislocating in the event of failure.

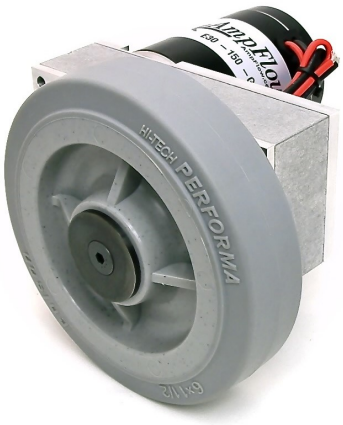


Fig. 25: 6 inches wheel on GearMotor

The reducer will be fixed to the chassis through a 1/8" aluminum plate. There will be a layer of lubricant between the aluminum plate and the square aluminum tube. The screws will go through a straight slot which will allow slight up-and-down movement. The displacement is absorbed by the rubber bumper pad on both sides of the reducer laying on the aluminum structure. It will ensure stability in the face of track irregularities. All the

fasteners will be in good condition and properly installed.

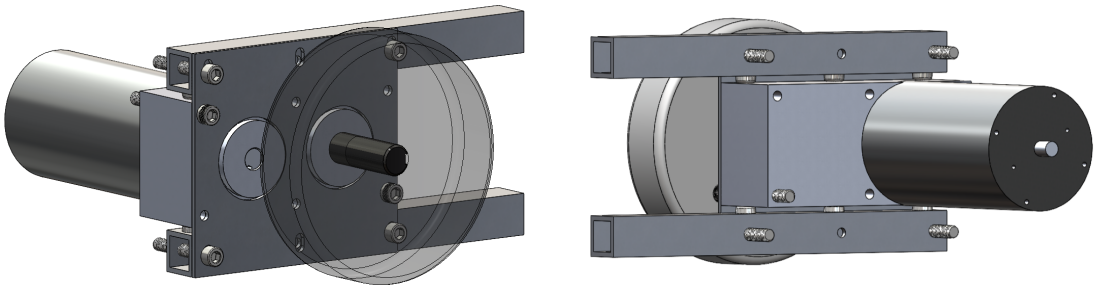


Fig. 26: Representation of the fixing methode of the speed reducer

Analysis

Also, the team did a measuring test to evaluate the impedance of each motor and see if the error was under 10%. Otherwise, the team must integrate a voltage synchronizer. The results came back positive with an oscillation from $190 - 200\mu H$. To continue, we verified if the vibrations coming from the nominal power were being tolerated by the motors. Fortunately, since the reducers are machined and the wheels are molded, their vibrations are minimized and their performance is maximized. Therefore, these tests were concluded.

Moreover, those speed reducers will be used as brakes as described in section 2.2.

2.8 Levitation

This year, and for the first time, PolyLoop is getting involved in the development of a levitation system. As working on a subject that can become such a complex design in the first year, our goal is not to levitate the entire pod but to start by building a first working levitation control system. The Research and Development team (previously known as the Physics team) is in charge of this project and now has a broader field of engineering branches associated to it. This first prototype will therefore help the team to learn how to collaborate efficiently and to start a new project. To make it more official, the deadline for the first working levitation system is set to February 2025. At the moment, our work focuses on finding the systems that have already been proposed in the bibliography and that could interest us such as: spinning permanent magnets arranged in a Halbach array, electromagnets or hybrid systems such as the one proposed by Swissloop on their Bertrand Piccard prototype. In order to implement them in future pods, the proposed systems will have to meet the constraints of the competition including: levitating on an aluminum track, being able to levitate the mass of the pod, etc. The necessary optimization of those systems to implement them in a pod on a track will be neglected for now. Nevertheless, the team is well aware that it will be, if not the biggest, a main challenge of their design. In the perspective of anticipation, we decided to participate in the Challenge 1 proposed by the competition, as a first opportunity to theoretically optimize our systems.

Manufacturing Process

For the manufacturing of the levitation systems, our team relies on the collaboration of mechanical, electrical, physics, civil, and computer engineering students who work together on designing prototypes, all the way from the mechanical structure and implementation in prototype concepts to microelectronic builds, physical designs, and hardware architectures. Although the team does not face as must of a critical deadline as the other branches of the design team, it still works with them under a shared perspective of anticipation to provide and receive systems such as PCBs. More concretely, the team's resources include those provided by Polytechnique Montreal, such as access to two active labs. Additionally, we have access to a professor's lab equipped with high-voltage infrastructure. Given these resources and the team's initial goal of creating a simple yet functioning system, the first prototype will primarily use components directly supplied by industry partners, sometimes facilitated through partnerships, allowing the team to establish a foundational system that can be optimized in future iterations.

Budget

Even though nothing has been purchased yet, a first but not precise budget has been dressed up for the entire team. However, the levitation part is one of our biggest interests.

Testing Process

The idea of making an initial, simple yet functional levitation system is also to learn from it and to test it efficiently. Given this objective, all prototypes created by the R&D team will be designed primarily to optimize testing. As we have done in the past with our small-scale linear induction motor, which sets a spinning aluminum wheel in motion, one potential design for a levitation system could involve levitating an aluminum plate with a time-varying mass distribution. For small-scale setups, the professor's lab mentioned earlier is the best resource available to the sub-team. The goal of these tests will be to characterize the systems, correlate these characterizations with theoretical models, and improve the systems. A strong emphasis will be placed on safety in all our work, especially when testing systems with electrical power.

2.9 Power Systems

placeins subcaption

Design Process

PolyLoop takes pride in making its own battery packs, thus allowing for safety, reliability and scalability to be customized. Last year, for the first time, we built and tested our 4 high voltage modules (48V, 16-26 A) powering the motors and the low voltage module (24V, 19.6A) powering all energy needs in embedded systems.

The linear induction motor is being built and characterized by our R&D team, after which the power systems team will build an appropriately sized battery.

In the meantime, the goal this year is to improve the overall power supply and distribution. This implies :

- Improving the conception of battery packs : bus bars shaping, spacers, charging mechanism;
- Develop the new casing structure of the modules : safety, fire-proofing, ventilation, isolation;
- Improve the way our battery cells are managed and monitored (new choice of BMS);
- Rethink the wiring system : wire gauge and color code, connectors, high voltage fuses and relays placement;
- Ensure a proper emergency system : down-sized manual disconnects, emergency stop replacement and developing a new additional emergency feature.

We will go through each of these poles in the following subsections. But in general, like last year, having four AmpFlow 48V motors to propel the capsule, we will also continue to power these four motors independently. We had four high-voltage modules, and one low-voltage module. For safety power-locking, each module has a manual switch system. All power systems are connected to an emergency stop button (E-stop).

Battery & battery casing

We use lithium-ion cells. Having already established our energy needs from last year, so far we plan to use the same battery configurations. Each high-voltage battery block has a theoretical usage autonomy of about 25 minutes, while the low-voltage battery has a theoretical usage autonomy of about 42 minutes. Regarding the spot welding of the different battery cells, we stay with a nickel, steel, copper alloy. Copper will allow us to have excellent conductivity, steel will provide robustness, while nickel will confer a good weld on the cells.

High voltage and low voltage configurations

We have **four high-voltage modules**, each consisting of a 48V (13s2p) battery with a continuous output of 16A and a peak of 26A.

The cell deemed to fit best our high-voltage needs is the Samsung INR18650-35E. The continuous usage autonomy from our last design was estimated at around 25 minutes. Below are its nominal specifications.

Item	Specification
3.1 Standard discharge Capacity	Min 3,350mAh - Charge : 0.5C(1,700mA), 4.2V, 0.02C(68mA) cut-off @RT - Discharge : 0.2C(680mA), 2.65V cut-off @RT *1C=3,400mA
3.2 Charging Voltage	4.2V
3.3 Nominal Voltage	3.60V
3.4 Charging Method	CC-CV (constant voltage with limited current)
3.5 Charging Current	Standard charge: 1,700mA For cycle life : 1,020mA
3.6 Charging Time	Standard charge: 4hours
3.7 Max. Charge Current	2,000mA (not for cycle life)
3.8 Max. Discharge Current	8,000mA (for continuous discharge) 13,000mA (not for continuous discharge)
3.9 Discharge Cut-off Voltage	2.65V
3.10 Cell Weight	50 g max
3.11 Cell Dimension	Height : Max. 65.25 mm Diameter: Max. Φ 18.55 mm
3.12 Operating Temperature (Cell Surface Temperature)	Charge : 0 to 45°C Discharge : -10 to 60°C
3.13 Storage Temperature	1 year : -20~25°C (1*) 3 months : -20~45°C (1*) 1 month : -20~60°C (1*)

Fig. 27: Samsung INR18650-35E nominal specifications (High Voltage)

Each pack is equipped with a BMS and supplying power through a fuse and a relay to power the PWM controllers and therefore the motors.

The conception of the **low-voltage battery** is compulsory to power the electronic components responsible for the motors control as well as the collection of real-time data. The energy needs of embedded systems are due to fluctuate depending on the final design chosen, but given the low-voltage battery of last year was over-sized, we can safely assume it will not exceed such energy needs. The configuration is a 7s2p, giving a battery of 24V and 19.6A continuous and 29.4A peak. Thus, keeping the same cell (assuming the need to be down-sized appropriately), we choose the Samsung INR21700-50E for its wide range in current and voltage capacity. Below are its nominal specifications.

Item	Specification
3.1 Standard discharge Capacity	Min 4,900mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 0.2C(980mA), 2.5V cutoff @ RT * 1C = 4,900mA
3.2 Rated discharge Capacity	Min 4,753mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 1C(4,900mA), 2.5V cutoff @ RT
3.3 Charging Voltage	4.2V
3.4 Nominal Voltage	3.6V
3.5 Charging Method	CC-CV (constant voltage with limited current)
3.6 Charging Current	Standard charge: 2,450mA
3.7 Charging Time	Standard charge: 3hours
3.8 Max. Charge Current	4,900mA (not for cycle life)
3.9 Max. Discharge Current	9,800mA (for continuous discharge) 14,700mA (not for continuous discharge)
3.10 Discharge Cut-off Voltage	2.5V
3.11 Cycle life	Capacity ≥ 3,802mAh @ after 500cycles (80% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, CCCV 0.05C(245mA) cut-off @ RT - Discharge: 1C(4,900mA), 2.5V cut-off @ RT
3.13 Recovery characteristics	Capacity recovery (after the storage) ≥ 4,278mAh (90% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Storage : 30 days @ 60℃ - Discharge : 1.0C(4,900mA) 2.50V cut-off @ RT
3.14 Cell Weight	69g max
3.15 Cell Dimension	Cell height : Max.70.80mm Diameter : Φ Max.20.25mm

Fig. 28: Samsung INR21700-50E nominal specifications (Low Voltage)

Like last year, we intend to put before each embedded system component a voltage controller (tension divider or else) to adjust the voltage to its nominal requirements.

To size our low-voltage and high battery packs, we selected lithium cells that match the desired characteristics using the lithium cell database on the "Kaispower" website. Subsequently, we compared them in terms of voltage, capacity, nominal discharge current, and price; testing various battery configurations. Given the preliminary choices made, these are the weights and sizes of our modules.

Weight

High-Voltage Module

According to the technical documentation of the "Samsung INR18650-35E" cell, it weighs 50g.

$$\text{Weight per module} = 26 \times 50 = 1300\text{g}$$
$$\text{Total H-V Weight} = 4 \times 1300 = 5200\text{g}$$

Low-Voltage Module

According to the technical documentation of the "Samsung INR21700-50E" cell, it weighs 69g each.

$$\text{Total L-V weight} = 14 \times 69 = 966\text{g}$$

Size

High-Voltage Module

According to the technical documentation of the "Samsung INR18650-35E" cell, each cell measures 18.55 mm in diameter and 65.25 mm in height.

$$\text{Length} = 13 \times 18.55 \text{ mm} = 24.115 \text{ cm}$$
$$\text{Width} = 2 \times 18.55 \text{ mm} = 3.71 \text{ cm}$$
$$\text{Height} = 65.25 \text{ mm} = 6.525 \text{ cm}$$

Low-Voltage Module

According to the technical documentation of the "Samsung INR21700-50E" cell, it measures 21mm in diameter and 70mm in height.

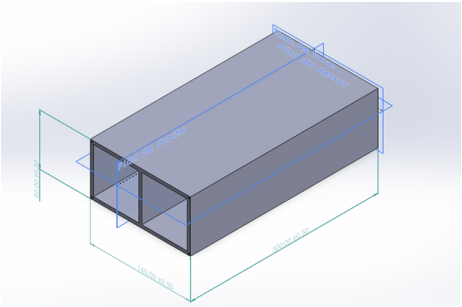
$$\text{Length} = 7 \times 21 \text{ mm} = 14.7 \text{ cm}$$
$$\text{Width} = 2 \times 21 \text{ mm} = 4.2 \text{ cm}$$
$$\text{Height} = 70 \text{ mm} = 7 \text{ cm}$$

Battery Casing

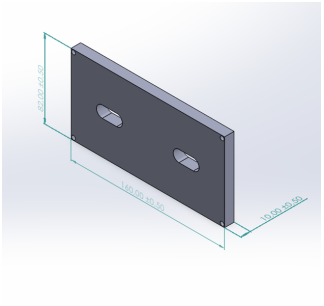
To enhance safety and performance, we are developing a new casing structure for the battery modules, focusing on fire-proofing, ventilation, and thermal isolation. This replaces the previous plastic spacers + heavy duty plastic wrap combination we used lasted year.

We first selected an aluminum case for its fire-resistant properties, lightweight durability, and excellent heat dissipation, which helps regulate internal temperatures under heavy load. Additionally, the casing will include an internal lining of aramid fiber sheets, a material known for its insulating properties. This aramid lining provides extra protection against heat and electrical conductivity, ensuring that the modules are isolated safely within the casing, even in demanding operating conditions.

Now, however, given time constraints and material availability, we opt to 3d print a case made of polyamid, a glass reinforced nylon filament. We thus modelized the casing design as show in the image.



29.1 Battery casing design



29.2 Cover battery casing design

Fig. 29: 3D models of the battery casing and its cover.

Battery management system

Last year, we used the Daly BMS, which performed well but lacked some of the high-end features we desired. This year, we're upgrading to a more versatile and customizable solution, the ENEPAQ TinyBMS s516, which offers improved reliability, customization, and monitoring capabilities.

The ENEPAQ TinyBMS s516 is designed for lithium-ion and LiFePO4 chemistries and supports a range of configurations, from 13s 48V to 7s 24V, with a continuous discharge current of 30A and a peak of 60A when mounted on a cool surface. Some of its key specifications include:

- **Cell Support**: Manages 4 to 16 cells with individual voltage monitoring, offering precise control with a resolution of 1mV.
- **Current, Voltage, and Temperature Management**: The BMS supports continuous and peak current tracking with protection features, including adjustable voltage and temperature cutoffs, ensuring safe operation across temperatures from -40°C to +85°C. We will particularly use its temperature tracking feature to monitor and manage battery temperatures in real time.
- **Balancing and Monitoring**: Features dissipative balancing at 150mA to maintain cell balance, along with data logging, event tracking, and customizable settings for discharge, charge, and recovery.
- **Interfaces**: Communication is supported via UART, USB, CAN, and Bluetooth, allowing for seamless integration with our monitoring systems.

The ENEPAQ TinyBMS s516, paired with the Battery Insider software, provides real-time monitoring of critical battery parameters, advanced data analysis capabilities, and lifetime statistics for enhanced troubleshooting and performance optimization. Additionally, its compact design (95 x 65 x 9 mm and 97g) makes it ideal for our applications where space is at a premium.

This upgrade will allow us to extend battery life, improve safety, and customize the BMS to align precisely with our project's power management requirements.

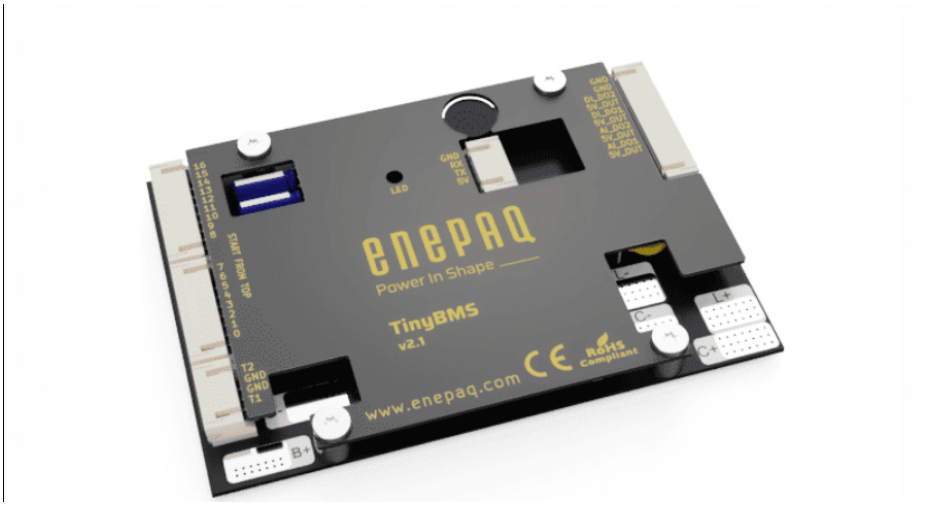


Fig. 30: ENEPAQ TinyBMS s516 Battery Management System

Wiring

We use 12AWG stranded cables, which can therefore support a current above our maximum of 25A at 48V. The color code is as follows:

- Red for low-voltage positive.
- Black for low-voltage negative.
- White with a red mark every 30 cm for high-voltage positive.
- White with a black mark every 30 cm for high-voltage negative.

To further distinguish between the low-voltage and high-voltage circuits, we use different connectors for each. For the low-voltage system, the connector used to connect with the battery is an XT60 connector, while for the connection between the high-voltage system and the battery, we use an XT90 connector.

Fuse

These components are part of our high-voltage system. The fuses are included to protect the high-voltage system against excessive current that could damage its various components.

We decided to size our fuses based on the PWM controller, as it is the component in our system that can handle the least current on the input side. It can handle a maximum of 25A at 48V.

Fuse Sizing: Given the previously mentioned information, we need a maximum current of 25A flowing through our circuit. However, for the fuse sizing, we should plan for a margin

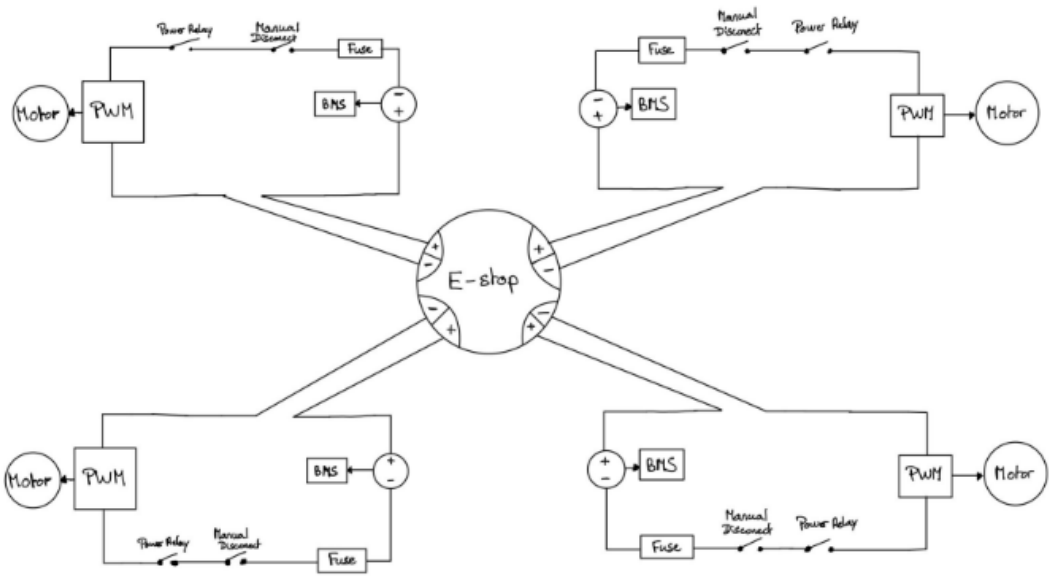


Fig. 31: Last year final cabling system

factor of approximately 1.25 between the desired and actual value.

$$\text{Fuse size} = 25A \times 1.25 = 31.25A$$

Thus, we can use a fuse with a nominal rating of 30A at 48V.

Relays

What is a relay?

A relay is an electromechanical component that allows one electrical circuit to be controlled by another, usually lower-power, control circuit. It acts like an automatic switch: when a current is applied to the control circuit (the coil), the relay closes or opens another circuit (the power circuit) capable of carrying a higher current. In our case, the relays are used to safely and controllably activate or deactivate power to the motors. They allow for a quick power cut when needed, for example, when using an emergency stop (E-stop) or a manual cut-off.

Use of Relays in Our System In our high-voltage propulsion system, relays play a key role in safely turning the motor power on and off. Here is their specific utility in our system:

Safety: Relays enable the motors' power supply to be cut in cases of overload, malfunction, or emergency stop, providing essential safety by quickly opening the power circuit when necessary.

Efficiency: Relays allow motor power to be controlled efficiently from a low-current control circuit, thereby avoiding exposing other components to high currents.

Isolation: In our system with four independent motors, the relays allow each motor module to be isolated, meaning a specific motor can be deactivated without affecting the others, facilitating maintenance and enhancing operational flexibility.

Emergency Stop System

In high-voltage battery systems, a reliable emergency stop (e-stop) mechanism is essential to ensure safety during operation and testing. An effective e-stop system allows for immediate power shutdown, reducing risks in unexpected situations and protecting both the equipment and personnel. Here are the key aspects our team is focusing on :

- **Resizing Manual Disconnects**: Previous e-stop systems were oversized, taking up excessive space in the wiring layout. The goal is to use manual disconnects better suited to our battery specifications, reducing cabling footprint. The new disconnect sizing is based on battery power with a 15% safety margin. Given the high-voltage battery (48V, 26A peak), the power requirement is calculated as:

  $$1,248\,\mathrm{W} \times 1.15 = 1,435.2\,\mathrm{W}$$

  New disconnects will be selected based on their ability to handle this power efficiently while optimizing size and integration.

- **Addition of an Extra Emergency Stop**: The e-stop needs to be visible and accessible at all times, but we cannot compromise the capsule's structure with additional holes. To address this, we plan to add a safety cord, similar to jet ski kill switches, that allows remote activation in an emergency. This cord, accessible even with the capsule's outer shell in place, will provide an extra layer of safety.

- **Safety Cord Challenges**: The key challenge is finding the optimal cord length. A longer cord could delay activation, while a shorter one may pose a tripping hazard. Balancing these factors is essential to ensure the e-stop is both effective and safe for the team.

2.10 Other (Auxiliary or Experimental)

Context

This year's research and experimental effort focus mainly on two projects : a levitation system and magnetic propulsion using a linear induction motors (LIMs). Since levitation has already been discussed in section 2.8, this section will mainly focus on magnetic propulsion. The research on LIMs follows the work conducted in fall 2023 and winter 2024, which centered on the design of an experimental linear induction motor. All the parts of the designed motor have been manufactured, and the remaining work is to assemble and test the motor, and that's what the team will focus on for the following year. Due to an error in the order, materials for the motor have been received twice. Therefore, the design has been change from a single sided LIM to a double-sided (DLIM). A more complete overview of the design and reasoning behind these choices will be discussed below.

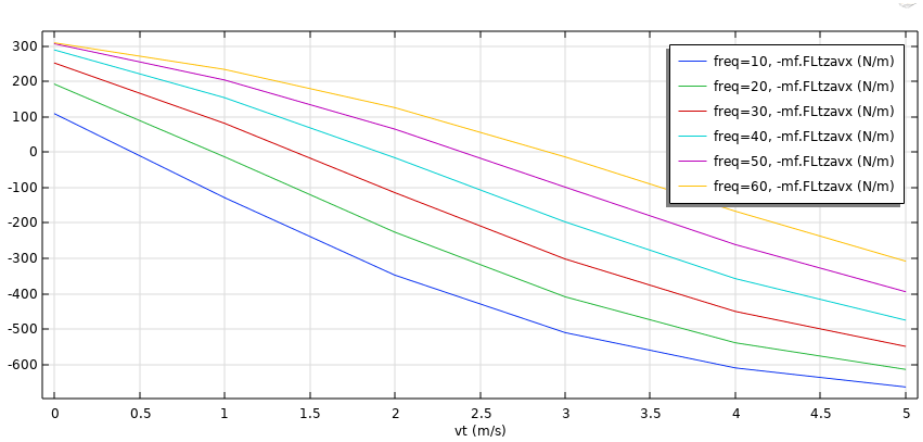
It is important to know that this motor is the second iteration of a LIM at PolyLoop. The first LIM, designed in 2021, aimed to provide 1g constant acceleration as the primary propulsion system for the entire pod. As a result, the motor designed was excessively heavy, large and demanded more power than our supply capacity could support. Although this motor was fully built and assembled, it was never implemented in the pod due to these limitations. It was also later found in 2023 that one of the coil was damaged, leading to its parts being repurposed for a miniature test LIM. The tests on this miniature LIM were extremely successful and have informed our approach for the design of our next LIM.

Design

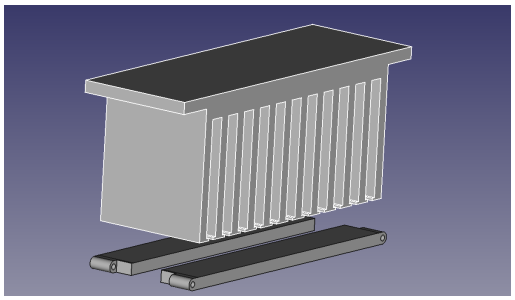
As mentioned earlier, the design of a single sided LIMs was completed during fall 2023 and winter 2024. Most design decisions were guided by simulations in COMSOL Multiphysics [1] and the assistance of a senior electrical engineer working for our partner *Delom Services*. The primary goal of this motor design was to expand on the miniature LIM's success while limiting the size and power to avoid issues encountered with our first iteration. Smaller goals were set and some constraints given by other teams were also given.

The power and battery team gave a limitation of 2000 W (100V/20A DC) as the maximum battery capacity they could manufacture. To limit the size of the motor, a maximum mass of [40kg] was settled, with a target mass of approximately 30kg. Within these constraints, the help and expertise of the senior engineer suggested us a 4 pole motor with 12 teeth. Further research concluded that concentric windings, while not as efficient as distributed windings, lead to a lighter and leaner design and were chosen for this motor.

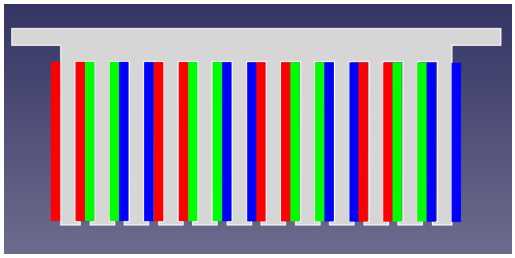
With the number of poles settled, simulations were conducted to determine the ideal current, number of turns per coil and the geometry using parametric sweeps and force analysis. Simulations results suggested 40 turns per coil and a current of 50 A (AC) per phase were ideal. Figure 32 illustrates simulated forces, phase configuration and a 3D model of our single sided design.



32.1



32.2



32.3

Fig. 32: (1) Forces (N/m) in function of pod speed (m/s) for multiple current frequencies (Hz). (2) 3D model of single sided LIM design. (3) Phase configuration of single sided motor design

Once the design completed and all components ordered from the manufacturer, the next step was to assemble and test the motor. However, as mentioned in the previous section, due to an error in the order, all material have been received twice without any additional fees. As a result, there was a possibility to build two motors and the team viewed this as an opportunity to implement and test double-sided linear induction motors. Advice from the senior engineer also suggested to us that this approach was best considering the situation. The engineer also recommended changing the phase configuration and adjusting the number

of turns to take in account the two motors. By considering his advice, we chose a distributed configuration in which each winding covers two teeth, as shown in Figure 33. The maximum number of turns was calculated using an LCR meter during the winding process and will be further discussed in the manufacturing section. The motors have not yet been tested, but are planned to be tested and functional before May.

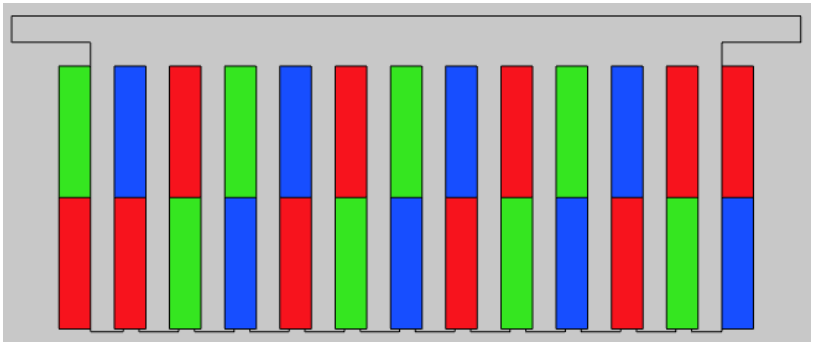


Fig. 33: Final phase configuration of our two linear inductions motors. 1st, 2nd and 3rd phase are shown in blue, red and green respectively

In addition to the motors themsleves, our team has been working on a test structure for the two LIMs. This structure is meant to be versatile, enabling testing both on our test-wheel and on the track. Its purpose is to facilitate testings on a smaller pod equipped with LIMs without time constraints or structural modification on the main pod. The structure is made of aluminum T-slotted extrusions, making it easy to assemble and adjust if necessary. Figure 34 shows the CAD model of our structure. The tests on the wheel will allow to experimentally characterize power consumption, forces and the frequency response of our DLIMs. Once these tests completed, the structure will allow for tests under real conditions on the track.

Manufacturing process

As mentioned in the previous section, manufacturing of the cores of the motors has been done by our partner, who also provided the necessary wires for coil assembly. The final manufacturing step involves determining the right phase configuration and assemble the coils. Due to the square shapes of the LIM, we were unable to use our partners' machinery for this step, so the coils were manufactured by hand. To do so, our team progressively winded the coils by adding only a few turns at a time. An LCR meter measuring the inductance was used to find the right number of turns (6) associated with the desired inductance. The overall budget of the two motors is estimated to be around 1500$ and the budget of the mini-pod structure to be around 500$.

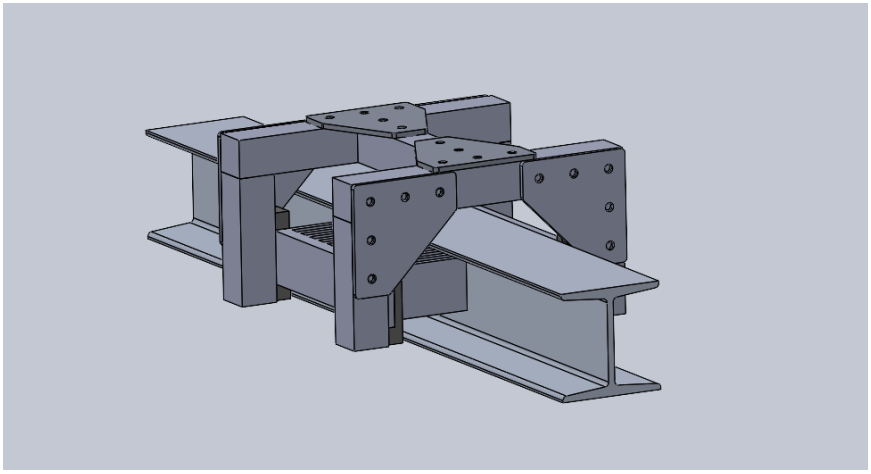


Fig. 34: CAD of the mini-pod structured used for testing the DLIMs

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[1] S. COMSOL AB, Stockholm, "Comsol multiphysiscs." [En ligne]. Disponible: www.comsol.com

[2] M. Hassani, D. Veysset, K. Nelson et C. Schuh, "In-situ observations of single micro-particle impact bonding," *Scripta Materialia*, vol. 145, p. 9–13, 10 2017.

[3] "Standard Test Methods for Tension Testing of Metallic Materials," American Association State, United States, Test Methods E8/E8M 22, juill. 2022. [En ligne]. Disponible: https://compass.astm.org/document/?contentCode=ASTM%7CE0008_E0008M-22%7Cen-US&proxycl=https%3A%2F%2Fsecure.astm.org&fromLogin=true

[4] C. Timperio, "Linear induction motor (lim) for hyperloop pod prototypes," *ETH Zürich*, 07 2018.

A Appendix - Design

Aeroshell appendices

Metric display of the mesh quality

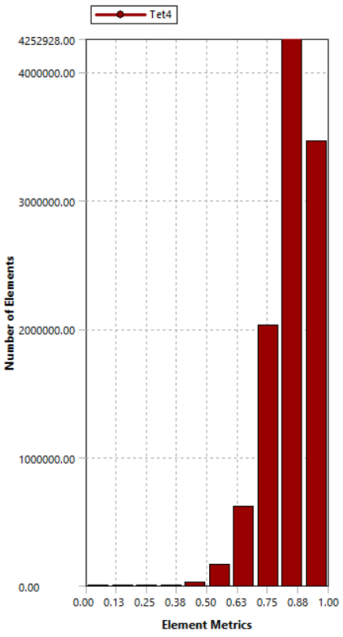

Fig. 35: Element Quality

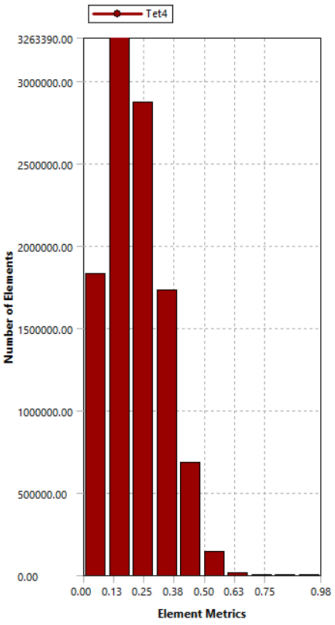

Fig. 36: Skewness

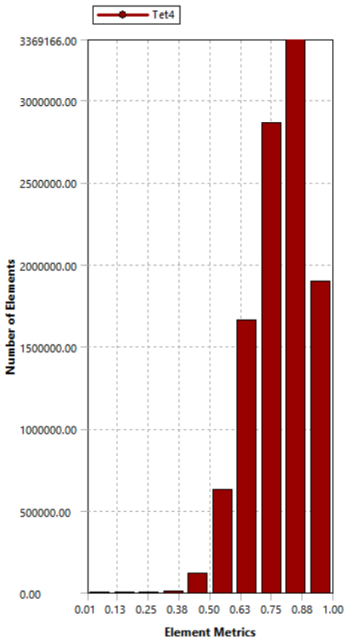

Fig. 37: Orthogonal Quality

y+ plot on the pod

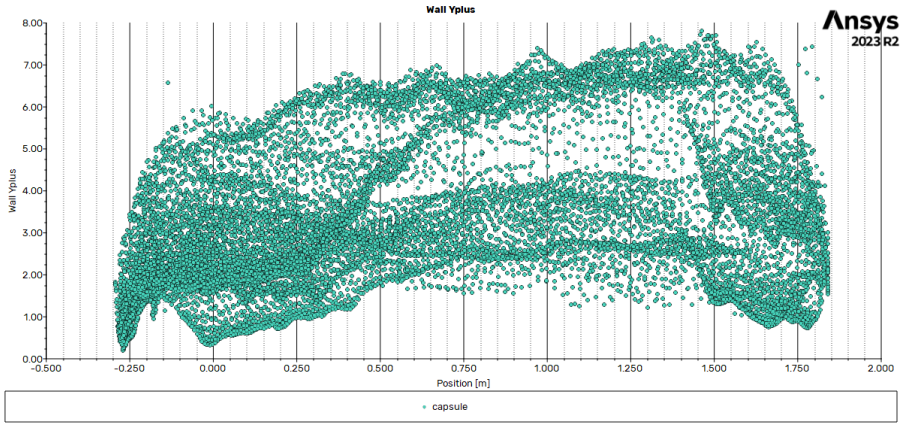

Fig. 38: y+ plot on the pod

y+ plot on the walls

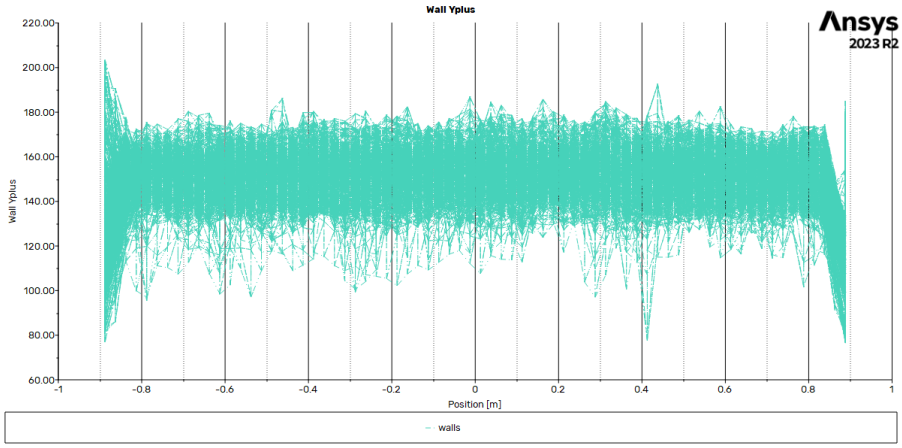

Fig. 39: y+ plot on the walls

Scaled residuals of the $k-\epsilon$ turbulent model used to obtain aerodynamics properties of the pod

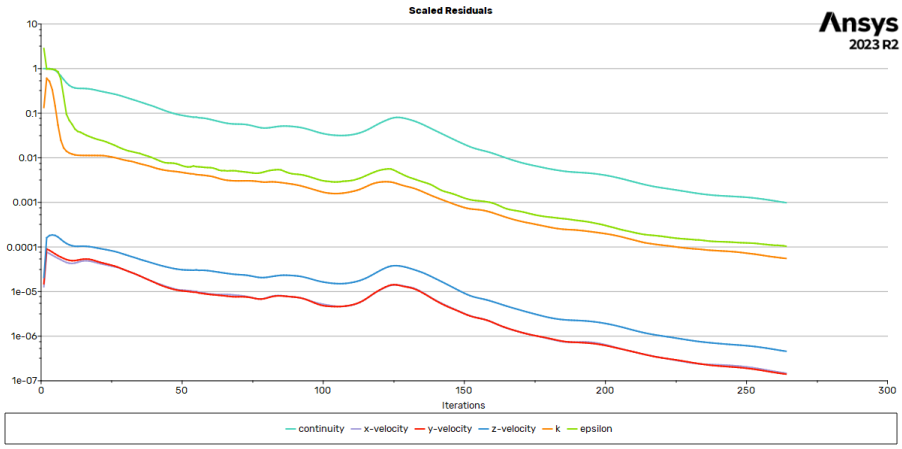

Fig. 40: Scaled residuals of the $k-\epsilon$ turbulent model

Plot of the drag coefficient over iteration

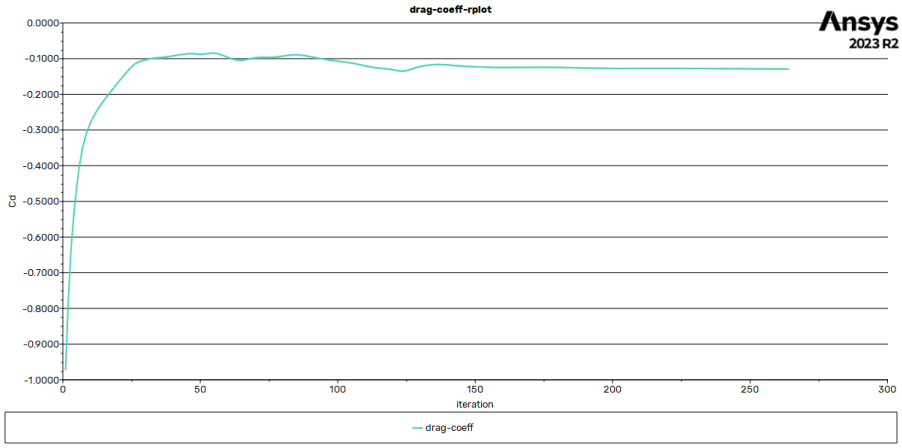

Fig. 41: Plot of the drag coefficient over iteration

Plot of the drag force over iteration

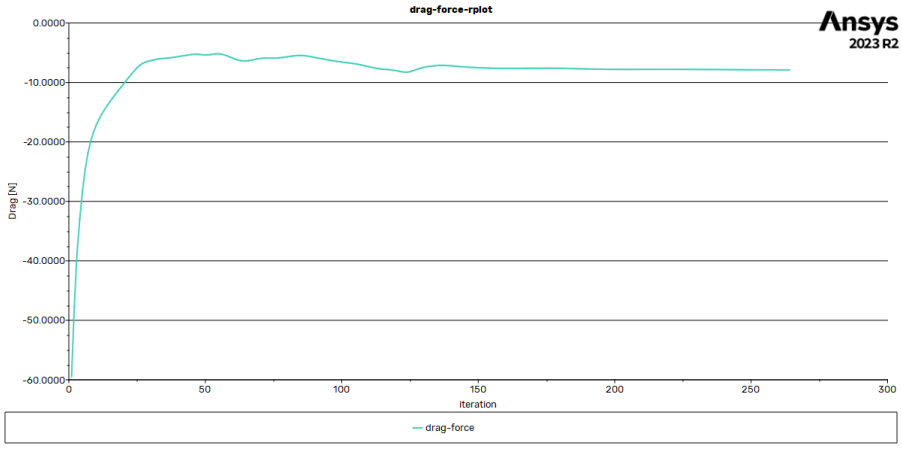

Fig. 42: Plot of the drag force over iteration

Plot of the lift force over iteration

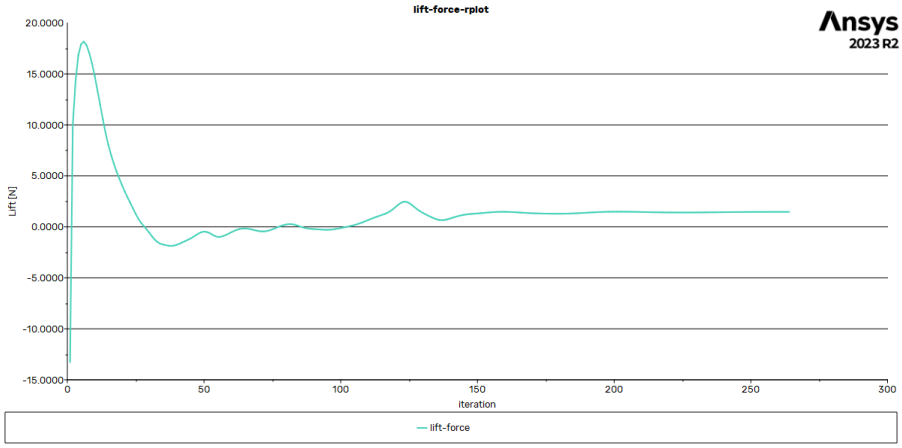

Fig. 43: Plot of the lift force over iteration

Plot of the lift coefficient over iteration

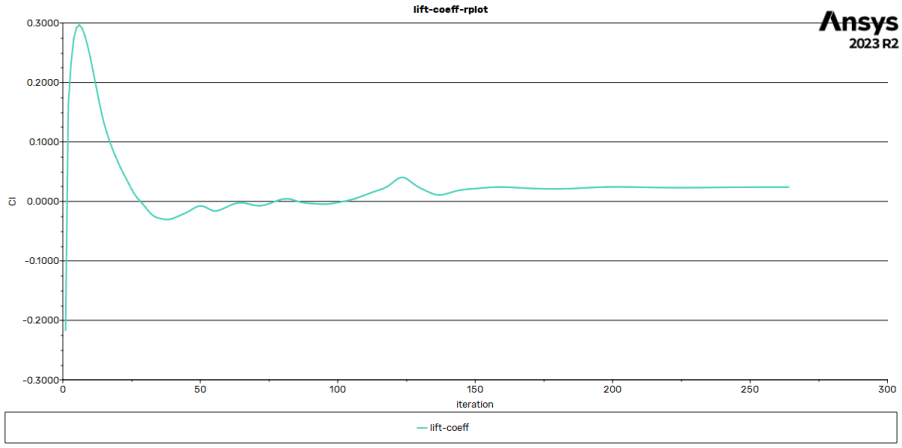

Fig. 44: Plot of the lift coefficient over iteration

Structures/Chassis appendices

Result of the structure modal analysis

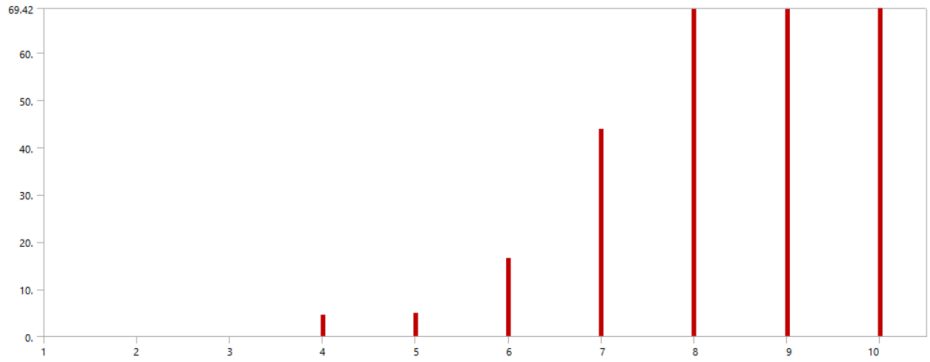

Fig. 45: 10 first natural frequency of the structure over a modal analysis (Hz)

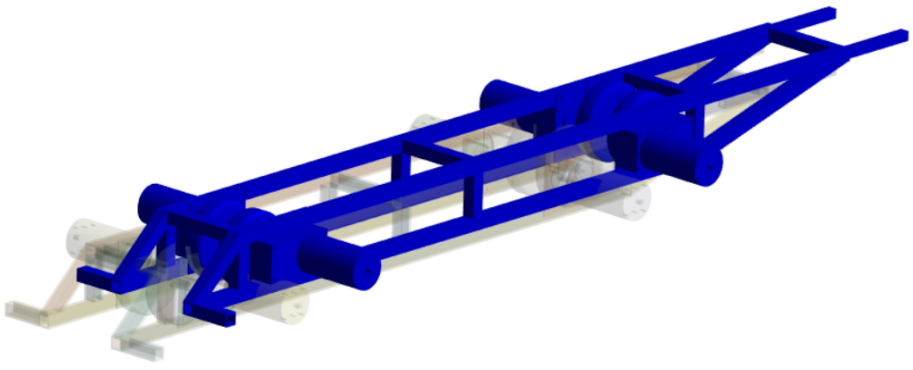

Fig. 46: Deformation at $0\,Hz$ (auto scale)

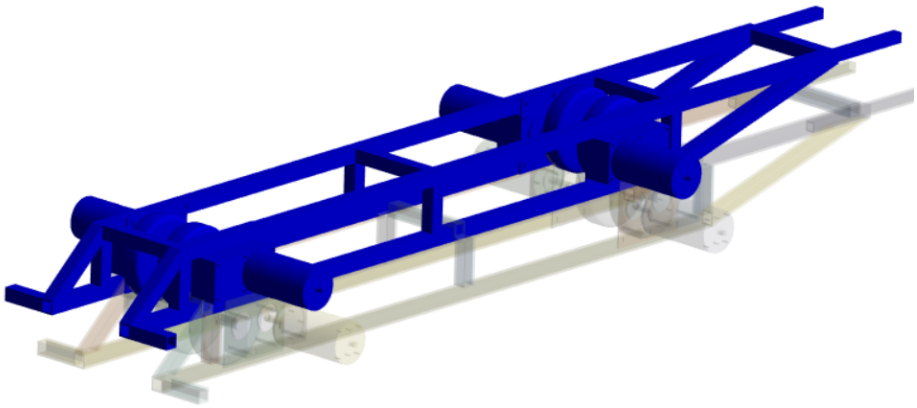

Fig. 47: Deformation at $0\,Hz$ (auto scale)

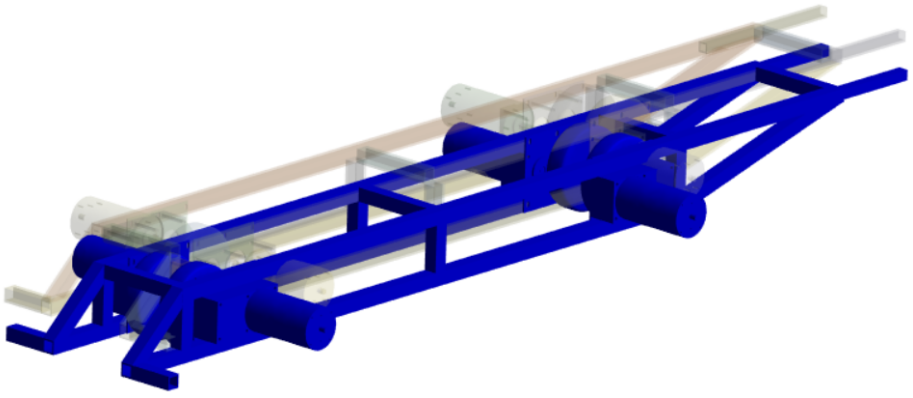

Fig. 48: Deformation at $2.4457e-004\,Hz$ (auto scale)

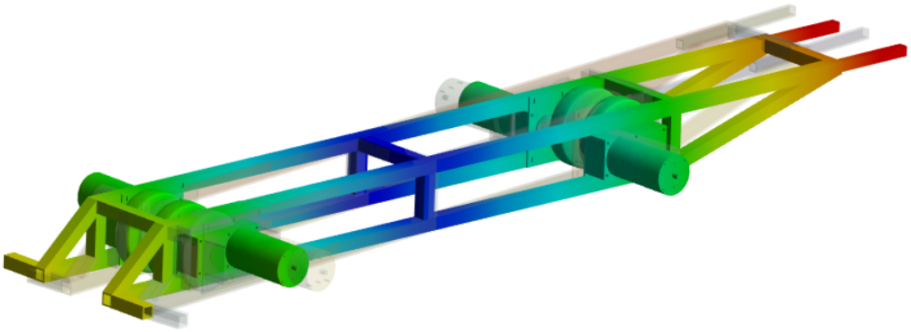

Fig. 49: Deformation at 4.4793 Hz (auto scale)

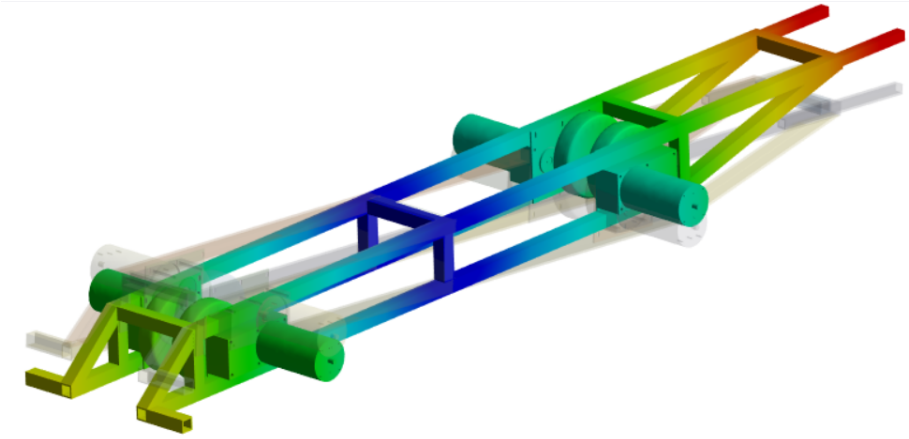

Fig. 50: Deformation at 4.9193 Hz (auto scale)

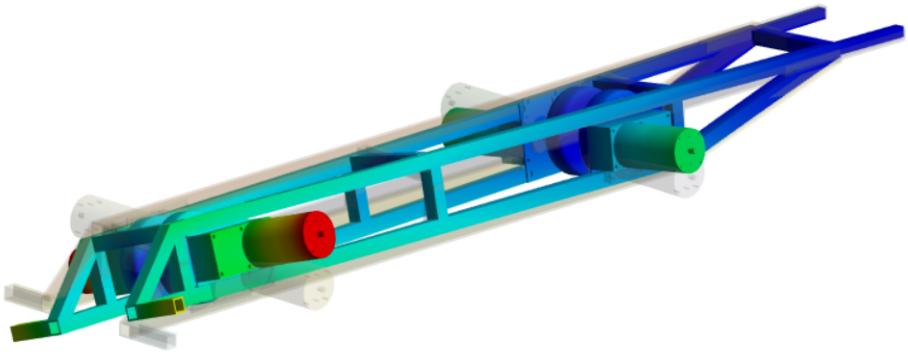

Fig. 51: Deformation at $16.612\,Hz$ (auto scale)

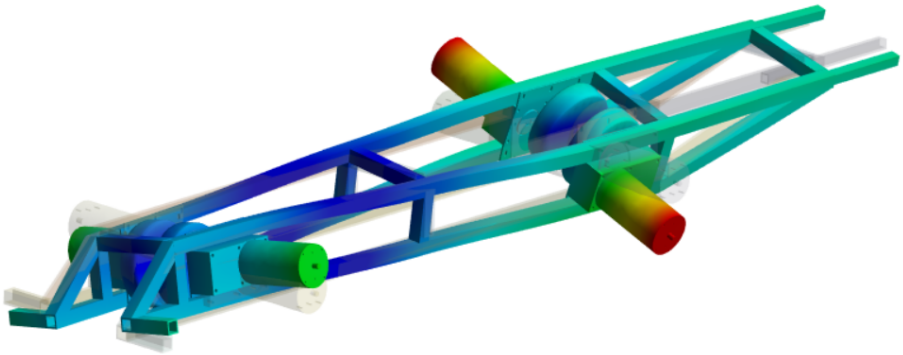

Fig. 52: Deformation at $43.835\,Hz$ (auto scale)

Propulsion appendices

E30-400-48-G4 manifacturer's performance table

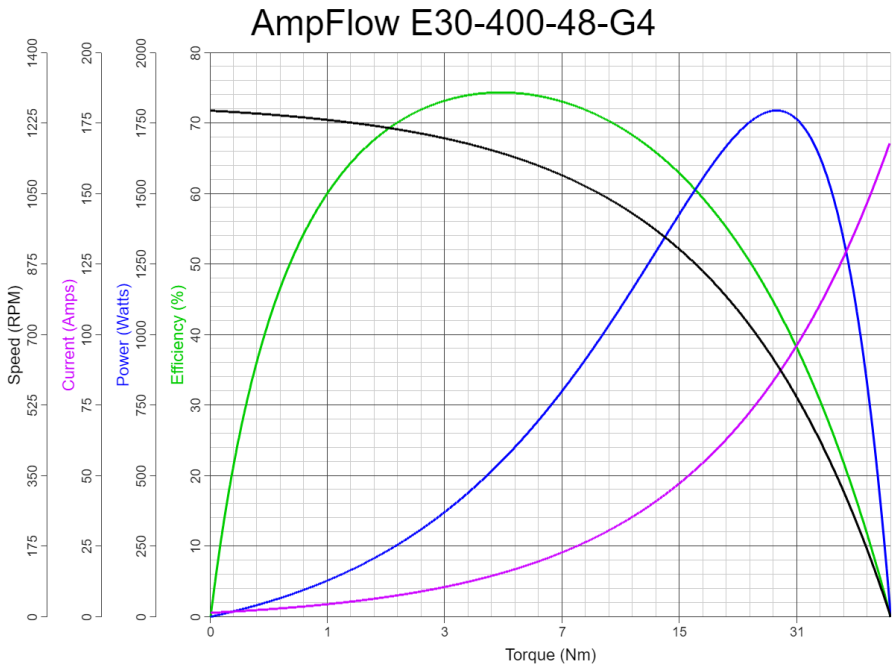

Fig. 53: E30-400-48-G4 manifacturer's performance table

Energy calculation for the propulsion needed

Energy calculation

To calculate the moment necessary to accelerate the capsule, it is necessary to use the method of Newton DCL, DCE on one of the wheels.

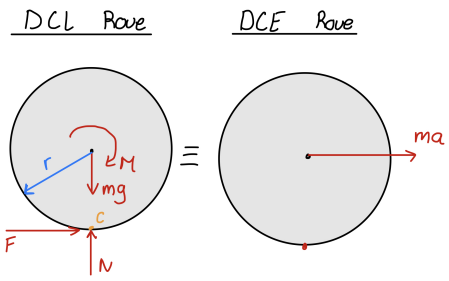

Fig. 54: DCL, DCE of one wheel

This way, by summing the moments on the instantaneous center of zero speed (ICZV),

we can determine the acceleration of the wheel.

$$\begin{aligned}\sum M_C = -M &= -m \cdot a \cdot r \\ 4.77 &= 25 \cdot a \cdot \frac{0.1524}{2} \\ a &= 2.5\end{aligned}$$

$a =$ acceleration $\left[\frac{m}{s^2}\right]$ $m =$ total mass on the wheel $[kg]$ $r =$ radius $[m]$

We know that the angular speed of the motor for a current of 16 Amps is $1150\,RPM$, we use this to calculate the nominal speed of the wheel.

$$\begin{aligned}v_{Roue} &= \omega \times R_{Rour} \\ &= \frac{1150 \cdot 2\pi}{60} \times \frac{0.1524}{2} \\ &= 9.17659\end{aligned}$$

$v_{Roue} =$ Velocity of the wheel $\left[\frac{m}{s}\right]$ $\omega =$ wheel rotation speed $\left[\frac{RAD}{s}\right]$ $R_{Rour} =$ radius of the wheel $[m]$

We find the time required by the capsule to reach this speed assuming that the acceleration is constant by the following relation;

$$\begin{aligned}a &= \frac{dv}{dt} \\ \int_0^t a dt' &= \int_0^v dv' \\ \int_0^t 2.5\ dt' &= \int_0^{9.18} dv' \\ 2.5\ t &= 9.18 \\ t &= 3.672\,\text{seconds}\end{aligned}$$

We now calculate the power needed to accomplish this acceleration by the relation of

the conservation of energy, neglecting the rotational kinetic energy of the wheel.

$$\begin{aligned} \text{Energy} \rightarrow P \cdot t &= \frac{1}{2} \cdot m \cdot v^2 \\ P &= \frac{m \cdot v^2}{2 \cdot t} \\ &= \frac{100 \cdot 9.18^2}{2 \cdot 3.627} \\ &= 1\,161\, Watt \end{aligned}$$

The theoretical power of an E30-400-48-G4 motor is $573\, Watt$. So, having 4 motors, the total power supplied is calculated:

$$\begin{aligned} P &= 573 \cdot 4 \\ &= 2292\, Watt \end{aligned}$$

Thus the power supplied is greater than the power required to accelerate the capsule to $9.18 m/s$, the E-30-400-48-G4 model is adequate.

Braking appendices

Analysis of the Hallback array

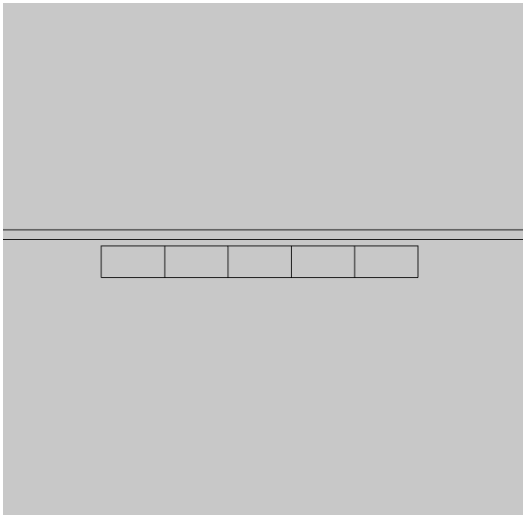

Fig. 55: Geometry used to simulate the effect of the Hallback array

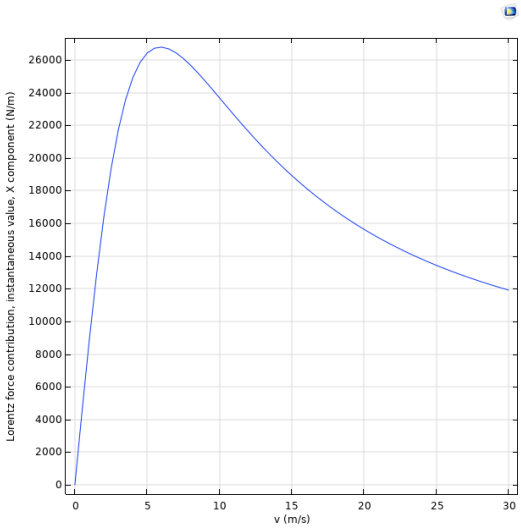

Fig. 56: Graph of the results of the simulation. The Lorentz force contribution is the braking force generated by the Hallback array. It is in function of the velocity v (m/s)

Acceleration compute from Newton second law

Calculation

We supose a constant mass m, constant forces and the momentum is define as $\vec{G} = m\vec{v}$.

$$\begin{aligned}\sum \vec{F} &= m\vec{a} = m\dot{\vec{v}} \\ &= m\frac{dv}{dt} = \frac{dm\vec{v}}{dt} \\ &= \frac{d\vec{G}}{dt}\end{aligned}$$

Then we distribute the dt and we integrate on both sides.

$$\begin{aligned}\sum \vec{F}\,dt &= d\vec{G} \\ \int_{t_0}^{t_1} \sum \vec{F}\,dt &= \int_{\vec{G}_0}^{\vec{G}_1} d\vec{G} \\ \sum \vec{F}(t_1 - t_0) &= \vec{G}_1 - \vec{G}_0 \\ \sum \vec{F} &= \frac{\vec{G}_1 - \vec{G}_0}{t_1 - t_0} \\ &= \frac{\Delta\vec{G}}{\Delta t}\end{aligned}$$

Where $\Delta\vec{G}$ is the change of momentum and Δt is the elapsed time. With this formula, we can compute the deceleration time knowing the masse of the pod m, the max velocity $\vec{v}_0$, and the total friction force.

Assuming total power failure, we sum the vector forces in the direction of the pod displacement $\vec{x}$. The only force acting is the friction force because the motors are in failure. With an initial speed $\vec{v}_0 = 9.14\,\vec{x}\,m/s$ and a total friction force $F_f = 159.052\,N$

$$\begin{aligned}\sum F|_x = -F_f &= \frac{\Delta G_x}{\Delta t} \\ \Delta t &= \frac{m\cancelto{0}{v_{1x}} - mv_{0x}}{-F_f} \\ &= \frac{-100\,[kg]\cdot 9.14\,[m/s]}{-159.052} \\ &= 5.7464\,s\end{aligned}$$

There the deceleration is $a = \frac{0-9.14}{5.7464} = -1.59\,m/s^2$.

Determination of the distance travelled in case of total power failure using only the speed reducer friction force

We hypothesize that the deceleration is constant and the only pod displacement direction is $\vec{x}$. The speed and deceleration could be describe as the rate of change of the over time such as;

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt}$$

Rearrange the termes to obtain:

$$dv = adt \rightarrow \int_{v_0}^{v_1} dv = a\int_{t_0}^{t_1} dt \rightarrow v_1 = v_0 + a(t_1 - t_0) \tag{i}$$

$$ds = vdt \rightarrow \int_{s_0}^{s_1} ds = \int_{t_0}^{t_1} \underbrace{(v_0 + a(t_1 - t_0))}_{\text{Eq}\,i}\,dt \rightarrow s_1 - s_0 = v_0 t + \frac{at^2}{2}$$

So the distance travelled is $\Delta s = 9.14 \cdot 22.3 + \frac{0.409 \cdot 22.3^2}{2} = 305.5\,m$